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Electrical and Computer Engineering

Modeling and Interface-Circuit Optimization for Piezoelectric Energy Harvesting: From Vibration Benchmarks to Low-Voltage Underwater Sources

Undergraduate Thesis

By: **Jodh Singh Nahal**

Supervisor: Prof. Lutz Lampe

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Abstract

This work investigates one continuous question across progressively more realistic circuit models: how the electrical interface of a piezoelectric energy harvester can be modeled, validated, and optimized when the available source changes from a conventional vibration-driven cantilever to an underwater acoustic transducer with very low terminal voltage. The first stage reproduced the impedance framework of Liang and Liao in MATLAB/Simscape for standard energy harvesting (SEH), parallel synchronized switch harvesting on inductor (P-SSHI), and series synchronized switch harvesting on inductor (S-SSHI). Rather than comparing only waveform shape or one output-power value, the simulation extracted the fundamental components of the simulated piezoelectric voltage and equivalent current, reconstructed the complex electrical impedance, and compared harvested and extracted power with the analytical expressions. At the final vibration benchmark, the maximum simulated load powers were approximately 0.49 mW for SEH, 1.75 mW for S-SSHI, and 2.18 mW for P-SSHI.

For a representative P-SSHI run at $R_{\text{load}} = 1\,\mathrm{M\Omega}$, the simulated harvested power was 2.09 mW, while Liang and Liao's harvested-power expression predicted 2.10 mW. The simulated fundamental extracted power was 3.83 mW, compared with 3.71 mW from the paper's analytical extracted-power expression. This agreement supported the use of the Simscape model as a basis for later circuit changes.

The ideal P-SSHI switch was then replaced with self-powered peak-detection circuitry based on the ESP-PSSHI topology of Zhang et al. The expected half-LC inversion time was 126.853 µs. The ideal switch closed for 124.916 µs with a measured delay of 40.525 µs from the equivalent-current zero crossing, while the BJT implementation closed for 133.849 µs but triggered 950.103 µs late. Its maximum harvested power was about 1.82 mW, showing that the duration of the LC inversion can be correct while the start time is still sufficiently delayed to reduce harvested power.

The same modeling workflow was then applied to an underwater acoustic equivalent source developed by Lutz Lampe. At 230 dB and 42 Hz, it gives an open-circuit amplitude of approximately 1.73 V. At 180 dB the same transducer representation gives only about

5.46 mV open circuit.

Once the circuit moved to LTspice, the goal changed from ideal switching to fast sweeps of real component SPICE models. A transformer/resonant matching network became particularly important. For a capacitive piezoelectric port, choosing the transformer primary magnetizing inductance near $L_p = 1/(\omega^2 C_p)$ cancels the piezoelectric susceptance at one operating frequency. At 42 Hz and 180 dB ($V_{\mathrm{oc}} \approx 5.46\,\mathrm{mV}$), PSSHI had no measurable power in LTspice ($< 10^{-15}$ W). Then testing a transformer it reached approximately 32 nW with an MBR0520L diode rectifier and matched primary inductance with 0.5 winding ratio, whereas a realistic transformer candidate with series resistance produced only about 1.15 nW. At 230 dB ($V_{\mathrm{oc}} \approx 1.73\,\mathrm{V}$), a low-loss, accurately tuned transformer model produced roughly 1.2 mW, while the compared P-SSHI model produced roughly 0.07 mW. However, small changes to L_p, C_p, or high winding resistance could reduce or reverse this advantage and finding a tranformer with large inductances (40 H @ 42 Hz) is challenging. Thus, transformer matching can outperform SSHI at selected operating points, but it is much more sensitive to frequency, capacitance, primary inductance, and series resistance.

The overall result is not a single universally best interface. P-SSHI gave the largest power in the validated vibration benchmark, while carefully tuned transformer matching was stronger in millivolt level underwater simulation where the rectifier forward voltage cannot be reached. The practical design therefore depends on source voltage, source impedance, frequency stability, component loss, and phase consistency when multiple piezoelectric elements are combined.

Acknowledgements

I would like to thank Prof. Lutz Lampe, Prof. Edmond Cretu and Dr. Masoud Ahmadi for guidance and for feedback throughout this research.

Contents

Chapter 1

Introduction

1.1 Motivation

Piezoelectric energy harvesters convert mechanical strain into electrical energy and can support low-power sensing systems where battery replacement is difficult. The useful output, however, depends on more than the piezoelectric material. The interface circuit determines when current is allowed to flow, how the piezoelectric capacitance is charged and discharged, how much energy is lost in switching and rectification, and the external load dependence.

This work began with a conventional vibration-driven benchmark because it provided published circuit equations, experimental parameters, and measured power curves against which a simulation could be checked. The same simulation framework was then extended to self-powered switching, underwater acoustic excitation, transformer matching, real component models, and multiple piezoelectric sources. These are successive simulations and circuit refinements directed at the same interface-design question.

1.2 Research Question

The research question used throughout the work is:

> *How can the electrical interface of a piezoelectric energy harvester be modeled and optimized so that useful power can be extracted under both conventional vibration excitation and extremely low-voltage underwater acoustic excitation?*

The question has two connected parts. First, the model must be accurate enough that analytical equations and time-domain simulation agree for a known benchmark. Second, it must demonstrate how to extract power at low source voltages, where ideal switches, fixed diode drops, and lossless magnetic elements are no longer valid assumptions.

1.3 Main Contributions

The main contributions of the work are:

1. a MATLAB/Simscape reproduction of the Liang–Liao SEH, P-SSHI, and S-SSHI impedance analysis, including direct comparison of waveforms, harvested power, and extracted power;
2. a timing-level and power analysis of a self-powered BJT P-SSHI detector;
3. application of equivalent underwater circuit representation at very low voltage levels;
4. a transition to LTspice for substantially faster component-level sweeps using real diode, transistor, inductor, and transformer models;
5. derivation and testing of a shunt-inductor/transformer impedance matching, including sensitivity to primary inductance, turns ratio, winding resistance, frequency, and piezo capacitance;
6. simulation of multiple piezoelectric elements with phase mismatch and alternative rectifier-combination strategies.
7. Choose circuit components to design and test a PCB.

Chapter 2

Analytical Model, MATLAB/Simscape Implementation, and Validation

2.1 Purpose and Validation Strategy

The first part of the work established a validated simulation framework before any underwater or component-level optimization was attempted. The vibration benchmark of Liang and Liao was useful because it provides the electromechanical equations, the SEH, P-SSHI, and S-SSHI interface equations, analytical expressions for equivalent impedance and power, and experimental power curves for comparison [1]. The objective here was not simply to reproduce those equations. The objective was to build the corresponding nonlinear interfaces in MATLAB/Simscape, extract the simulated waveforms, independently calculate the quantities used in the analytical model, and then determine whether the two approaches agreed.

2.2 Base-Excited Piezoelectric Model

2.2.1 Coupled electrical equations

Liang and Liao represent a base-excited piezoelectric harvester using a single-degree-of-freedom mechanical model coupled to the piezoelectric voltage v_p and current i_p [1]. The mechanical equations may be mapped into an equivalent electrical series RLC branch,

$$v_{eq}(t) = -\frac{M}{\alpha_e}\ddot{y}(t), \tag{2.1}$$

$$i_{eq}(t) = \alpha_e \dot{x}(t), \tag{2.2}$$

$$L = \frac{M}{\alpha_e^2}, \tag{2.3}$$

$$R = \frac{D}{\alpha_e^2}, \tag{2.4}$$

$$C = \frac{\alpha_e^2}{K + K_p}. \tag{2.5}$$

The mechanical dynamics are therefore represented by a series L–R–C branch driven by v_{eq}, while C_p and the harvesting circuit form the electrical part. In the frequency domain,

$$V_{eq}(j\omega) = \left(j\omega L + R + \frac{1}{j\omega C} + Z_{\text{elec}}(j\omega) \right) I_{eq}(j\omega), \tag{2.6}$$

with

$$Z_{\text{elec}}(j\omega) = \frac{V_p(j\omega)}{I_{eq}(j\omega)}. \tag{2.7}$$

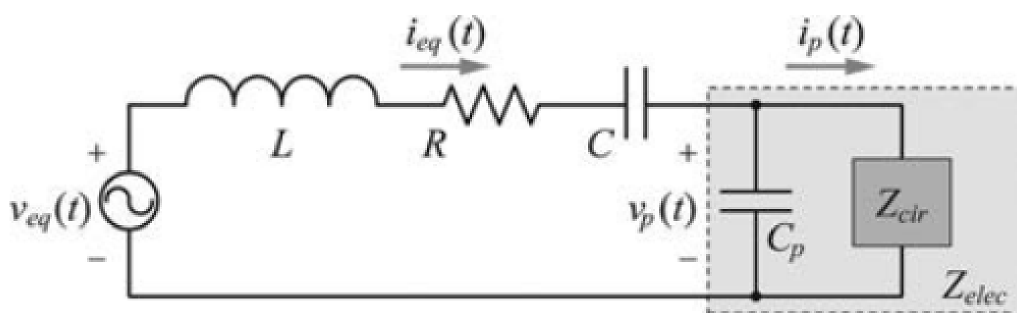


Figure 2.1: Base-excited piezoelectric circuit model used as the starting point for the Simscape model, reproduced from Liang and Liao [1].

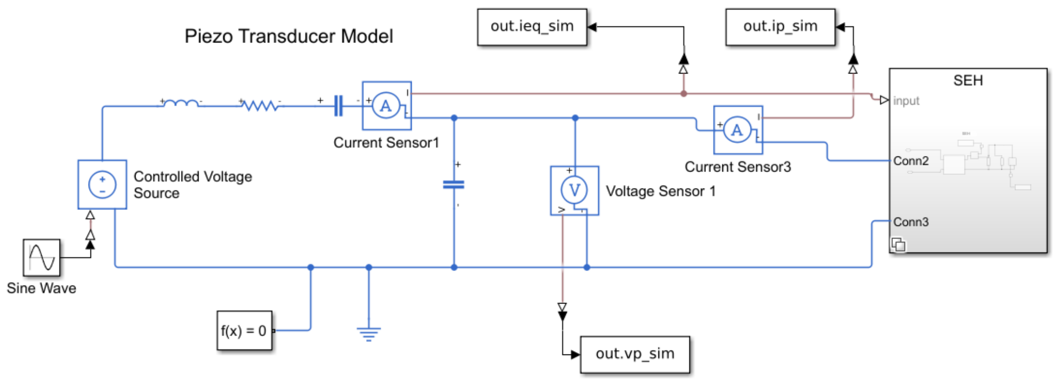


Figure 2.2: Simscape model. The interface circuit subsystem on the right can be set to SEH, P-SSHI or S-SSHI. Also contains voltage/current test points.

2.2.2 Benchmark parameters and source amplitude

The benchmark values used in the Simscape model were

Table 2.1: Vibration-benchmark parameters used for the MATLAB/Simscape reproduction.

Quantity	Symbol	Value
Excitation frequency	f	42 Hz
RMS base acceleration	$\ddot{Y}_{\mathrm{rms}}$	$10\,\mathrm{m/s^2}$
Equivalent inductance	L	31 kH
Equivalent resistance	R	1 MΩ
Equivalent capacitance	C	448 pF
Piezoelectric capacitance	C_p	34.69 nF
Force–voltage factor	α_e	4.75×10^{-4}
Storage capacitor	C_{rect}	1 µF
Bridge drop used in benchmark validation	V_F	approximately 1 V total
P-SSHI/S-SSHI inversion inductor	L_i	47 mH
Voltage inversion factor	γ	-0.7

The RMS acceleration was converted to peak acceleration,

$$\ddot{Y}_{pk} = \sqrt{2}\,\ddot{Y}_{\mathrm{rms}}, \tag{2.8}$$

and because $M/\alpha_e = L\alpha_e$, the equivalent source amplitude used in the Simscape model was

$$V_{eq,pk} = L\alpha_e \ddot{Y}_{pk}. \tag{2.9}$$

For the values above this gives approximately 208 V. This large internal equivalent voltage should not be confused with the piezoelectric terminal voltage; it appears across the complete equivalent network.

The MATLAB implementation was

```
Y_rms_accel = 10;
y_peak_accel = sqrt(2) * Y_rms_accel;

L = 31e3;
R = 1e6;
C = 448e-12;
Cp = 34.69e-9;
ae = 4.75e-4;

VSrc_eq_amp = L * ae * y_peak_accel;
```

2.2.3 Open-circuit voltage check

Before connecting any interface, the open-circuit terminal voltage was checked from the equivalent circuit. With

$$Z_m = R + j\omega L + \frac{1}{j\omega C}, \tag{2.10}$$

$$Z_{C_p} = \frac{1}{j\omega C_p}, \tag{2.11}$$

the open-circuit terminal voltage is

$$V_{oc} = \left| V_{eq} \frac{Z_{C_p}}{Z_m + Z_{C_p}} \right|. \tag{2.12}$$

The corresponding code was

```
w = 2*pi*f;
Zseries_oc = R + 1j*w*L + 1/(1j*w*C);
ZCp_oc = 1/(1j*w*Cp);

openCircuitVoltage = abs( ...
    VSrc_eq_amp * ZCp_oc / (Zseries_oc + ZCp_oc) );
```

The simulated open-circuit waveform was also measured directly and its fundamental component was extracted. The measured fundamental amplitude used in the later validation was approximately 20 V. Liang and Liao's Eq. 44 can be rearranged to infer the equivalent source from the measured open-circuit voltage,

$$V_{eq} = \frac{\left|R + j(X_L + X_C + X_{C_p})\right|}{|jX_{C_p}|} V_{oc}, \tag{2.13}$$

where $X_L = \omega L$, $X_C = -1/(\omega C)$, and $X_{C_p} = -1/(\omega C_p)$.

2.3 How the Simulated Waveforms Were Processed

2.3.1 Steady-state interval and final cycle

Each Simscape run was allowed to settle before any analytical comparison was made. A steady-state interval was retained for load-power averaging, and one complete final cycle was isolated for waveform, phase, impedance, and switching calculations:

TODO: what is vp

```
idx_ss = t >= t_start_ss;
t_ss = t(idx_ss);
vp_ss = vp(idx_ss);
vstore_ss = vstore(idx_ss);
ieq_ss = ieq(idx_ss);

T = 1/f;
t2 = t_ss(end);
t1 = t2 - T;
idx_cycle = t >= t1 & t <= t2;

t_c = t(idx_cycle);
vp_c = vp(idx_cycle);
ieq_c = ieq(idx_cycle);
vstore_c = vstore(idx_cycle);
tau = t_c - t_c(1);
```

This prevented startup charging of C_{rect} from being counted as steady harvested power.

2.3.2 Fundamental extraction from the actual simulated signals

A key assumption in Liang and Liao is that the equivalent current can be regarded as a perfect sine wave,

$$i_{eq}(t) = I_0 \sin(\omega t), \tag{2.14}$$

and that only the fundamental harmonic of v_p materially affects the electromechanical dynamics [1]. In the simulation, however, neither the amplitude nor the phase was inserted from the analytical model. Both were obtained from the actual final-cycle waveforms.

Each simulated signal $y(t)$ is fitted to a single-harmonic model using the method of least squares:

$$y(t) \approx a \sin(\omega t) + b \cos(\omega t) \tag{2.15}$$

The fundamental amplitude and phase are then

$$A = \sqrt{a^2 + b^2}, \qquad \phi = \operatorname{atan2}(b, a), \tag{2.16}$$

and the reconstructed fundamental is

$$y_F(t) = A \sin(\omega t + \phi). \tag{2.17}$$

The exact calls used in the simulation were

```
[vp_F, VpF_amp, VpF_phase] = ...
   fundamentalComponent(t_c, vp_c, f);

[ieq_F_raw, I0_raw, ieq_phase_raw] = ...
   fundamentalComponent(t_c, ieq_c, f);
```

and inside `fundamentalComponent`:

```
y = y - mean(y, "omitnan");
w = 2*pi*f;

A = [sin(w*t), cos(w*t)];
coeff = A \ y;
a = coeff(1);
b = coeff(2);

amp = hypot(a, b);
phase = atan2(b, a);
```

```
yF = amp * sin(w*t + phase);
```

The phasors were then formed from the fitted amplitudes and phases,

$$V_{p,F} = V_{p,F} e^{j\phi_v}, \tag{2.18}$$

$$I_{eq,F} = I_0 e^{j\phi_i}, \tag{2.19}$$

and the simulated equivalent impedance was

$$Z_{\text{elec,sim}} = \frac{V_{p,F}}{I_{eq,F}}. \tag{2.20}$$

This impedance is decomposed later and used to calculate the numerical power result. Thus, the analytical paper model was used as a validation target, while the quantities being compared against it came from the nonlinear Simscape waveforms themselves [9].

2.3.3 Rectified voltage and power calculated from simulation

The average storage voltage (the voltage across C_{rect}) was calculated over the retained steady-state interval,

$$\overline{V}_{store} = \frac{1}{t_b - t_a} \int_{t_a}^{t_b} v_{store}(t)\, dt, \tag{2.21}$$

and the sum of $\overline{V}_{store}$ and the forward voltage drop of the bridge rectifier V_f (twice the diode forward voltage) was used to get

$$V_{rect} = \overline{V}_{store} + V_F. \tag{2.22}$$

which is used in further equations. The implementation was

```
Vstore_avg = trapz(t_ss, vstore_ss) / ...
            (t_ss(end) - t_ss(1));

Vrect_sim = Vstore_avg + VF_bridge;
```

Three power quantities were calculated independently.

First, useful load power was evaluated from the full steady state storage-voltage waveform rather than from one sample:

$$P_{load} = \left\langle \frac{v_{store}^2(t)}{R_{load}} \right\rangle, \tag{2.23}$$

The code was

```
Pload_inst = vstore_ss.^2 ./ Rload;
Pload_avg = trapz(t_ss, Pload_inst) / ...
           (t_ss(end) - t_ss(1));
```

Second, total electrical power extracted from the equivalent source was calculated directly over the final cycle,

$$P_{\Delta,time} = \langle v_p(t) i_{eq}(t) \rangle \,, \tag{2.24}$$

using

```
Pdelta_time = trapz(t_c, vp_c .* ieq_c_eff) / ...
             (t_c(end) - t_c(1));
```

TODO: why use 1 cycle, use ss

Third, because the paper's impedance model is a fundamental-harmonic model, a directly comparable fundamental extracted power was calculated as

$$P_{\Delta,F} = \frac{1}{2}\Re\{Z_{\text{elec,sim}}\} I_0^2. \tag{2.25}$$

The comparison with paper equations and simulation was made after extracting I_0, phase, and V_{rect} from the simulation itself. The code first used the fitted I_0 to obtain V_{oc}, then used the simulated V_{rect} to calculate $\widetilde{V}_{rect}$ and θ, and finally evaluated all the paper equations including piezo voltage, impedance and power.

2.4 Standard Energy Harvesting (SEH)

2.4.1 SEH model

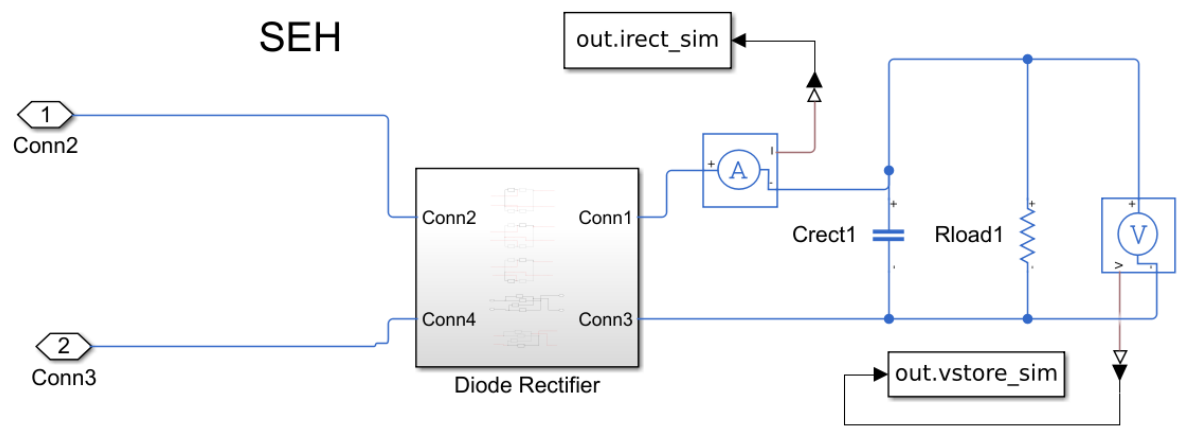


Figure 2.3: Internals of Simscape subsystem on the right from Figure 2.2 for SEH interface circuit containing rectifier and smoothing capacitor C_{rect} with R_{load}.

2.4.2 SEH waveform and impedance equations

For SEH, the full-wave bridge clamps v_p to $\pm V_{rect}$ when the rectifier conducts. During the blocked interval, the equivalent current charges or discharges C_p. With the sine-wave equivalent current of Eq. (2.14), the open-circuit voltage magnitude is

$$V_{oc} = \frac{I_0}{\omega C_p}. \tag{2.26}$$

Define the normalized rectified voltage

$$\widetilde{V}_{rect} = \frac{V_{rect}}{V_{oc}}. \tag{2.27}$$

The rectifier blocked angle θ is related to V_{rect} by

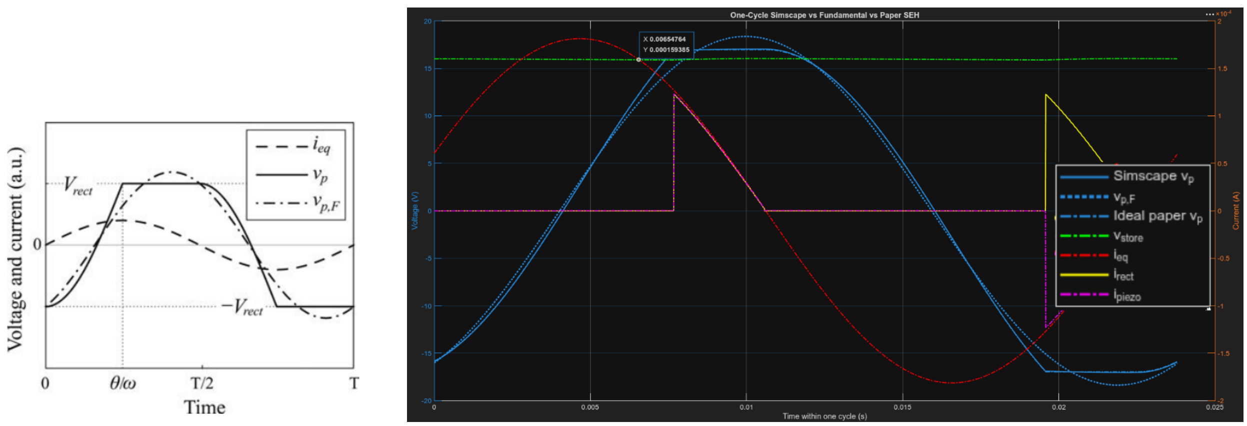
$$\cos\theta = 1 - 2\widetilde{V}_{rect}. \tag{2.28}$$

The paper's one-cycle SEH voltage waveform is then

$$v_p(t) = \begin{cases} V_{oc}[1-\cos(\omega t)] - V_{rect}, & 0 \le \omega t < \theta, \\ V_{rect}, & \theta \le \omega t < \pi, \\ V_{rect} - V_{oc}[1+\cos(\omega t)], & \pi \le \omega t < \pi + \theta, \\ -V_{rect}, & \pi + \theta \le \omega t < 2\pi. \end{cases} \tag{2.29}$$

TODO

This allowed a direct overlay of the simulated $v_p(t)$, the paper waveform, the simulated equivalent current, and its fitted fundamental on the same final cycle.



(a) Paper theoretical curves.

(b) Simscape simulated SEH curves.

Figure 2.4: Simulation results look identical to paper graph and numerical paper equations waveform. The measured solid blue V_p from simulation is directly matched with the dotted blue ideal paper V_p, Eq. (2.29) and the red I_{eq} curve being out of phase like in the paper graph. Figure (a) taken from [1, Fig. 3].

The fundamental component obtained analytically from this Eq. (2.29) waveform is

$$v_{p,F}(t) = \frac{I_0}{2\pi\omega C_p}\left([\sin(2\theta) - 2\theta]\cos(\omega t) + 2\sin^2\theta\sin(\omega t)\right), \tag{2.30}$$

and therefore

$$Z_{\text{elec,SEH}} = \frac{1}{\pi\omega C_p}\left[\sin^2\theta + j(\sin\theta\cos\theta - \theta)\right]. \tag{2.31}$$

To distinguish useful harvested power from circuit loss, the real part of Z_{elec} is decomposed into a harvesting resistance R_h and a dissipative resistance R_d, while the imaginary part is represented by X_E. With

$$\widetilde{V}_F = \frac{V_F}{V_{oc}}, \tag{2.32}$$

Liang and Liao give for SEH

$$R_d = \frac{4}{\pi\omega C_p}\widetilde{V}_F(1 - \widetilde{V}_{rect}), \tag{2.33}$$

$$R_h = \frac{4}{\pi\omega C_p}(\widetilde{V}_{rect} - \widetilde{V}_F)(1 - \widetilde{V}_{rect}), \tag{2.34}$$

$$X_E = \frac{1}{\pi\omega C_p}(\sin\theta\cos\theta - \theta). \tag{2.35}$$

Therefore,

$$Z_{\text{elec}} = (R_d + R_h) + jX_E. \tag{2.36}$$

For a representative $1\,\text{M}\Omega$ validation run, the simulated and analytical equivalent impedances were approximately

$$Z_{\text{elec,sim}} \approx (16.6 - j98.5)\,\text{k}\Omega, \tag{2.37}$$

$$Z_{\text{elec,paper}} \approx (16.6 - j98.0)\,\text{k}\Omega. \tag{2.38}$$

2.4.3 SEH harvested and extracted power

After obtaining R_h, R_d, and X_E, the analytical harvested and extracted powers were calculated with Liang and Liao's Eqs. 42 and 43:

$$P_h = \frac{V_{eq}^2}{2}\frac{R_h}{(X_L + X_C + X_E)^2 + (R + R_d + R_h)^2}, \tag{2.39}$$

$$P_\Delta = \frac{V_{eq}^2}{2}\frac{R_h + R_d}{(X_L + X_C + X_E)^2 + (R + R_d + R_h)^2}. \tag{2.40}$$

At a representative $1\,\text{M}\Omega$ SEH point, the comparison was approximately

Table 2.2: Representative SEH power comparison at $R_{load} = 1\,\text{M}\Omega$.

Quantity	Power
Simscape harvested/load power, Eq. (2.23)	0.255 mW
Eq. 42 harvested power	0.258 mW
Time-domain extracted power, Eq. (2.24)	0.321 mW
Fundamental extracted power, Eq. (2.25)	0.282 mW
Eq. 43 extracted power	0.274 mW

The load-power and Eq. 42 results are close, and the fundamental extracted-power result is especially useful for comparison with Eq. 43 because both quantities use the same fundamental-harmonic representation.

2.4.4 Load-Resistance Sweep Method

The load sweep was performed by changing R_{load} while changing the simulation duration. The output time constant changes as

$$\tau_{RC} = R_{load} C_{rect}, \tag{2.41}$$

so high-resistance loads require a longer settling time.

The script therefore generated a logarithmic load vector and adjusted the simulation stop time according to the output time constant:

```
Rload_list = logspace(4, 7, 20);

for k = 1:numel(Rload_list)
    Rload_k = Rload_list(k);
    tauRC = Rload_k * Crect;

    stopTime_k = max(minStopTime, tauMultiplier*tauRC);
    stopTime_k = min(stopTime_k, maxStopTime);

    t_start_ss_k = ssStartFraction * stopTime_k;

    sweepResults(k) = runOnePEH( ...
        ..., Rload_k, ..., ...
        stopTime_k, t_start_ss_k, ...);
end
```

For each load value the same steady-state and fundamental extraction described above was repeated. The optimum was selected using the simulated average dc load power:

```
validP = isfinite(sweepTable.Pload_avg_W);

[Pmax, idxLocal] = ...
    max(sweepTable.Pload_avg_W(validP));

validIdx = find(validP);
idxOpt = validIdx(idxLocal);

Rload_opt = sweepTable.Rload_ohm(idxOpt);
Pload_opt = sweepTable.Pload_avg_W(idxOpt);
```

The sweep stored, for every R_{load}, the simulated harvested power, time-domain extracted power, fundamental extracted power, analytical Eq. 42 power, analytical Eq. 43 power, V_{store}, V_{rect}, $\widetilde{V}_{rect}$, $Z_{\text{elec,sim}}$, $Z_{\text{elec,paper}}$, R_d, R_h, and X_E. This made the load sweep a full model-to-model comparison rather than only an output-power search.

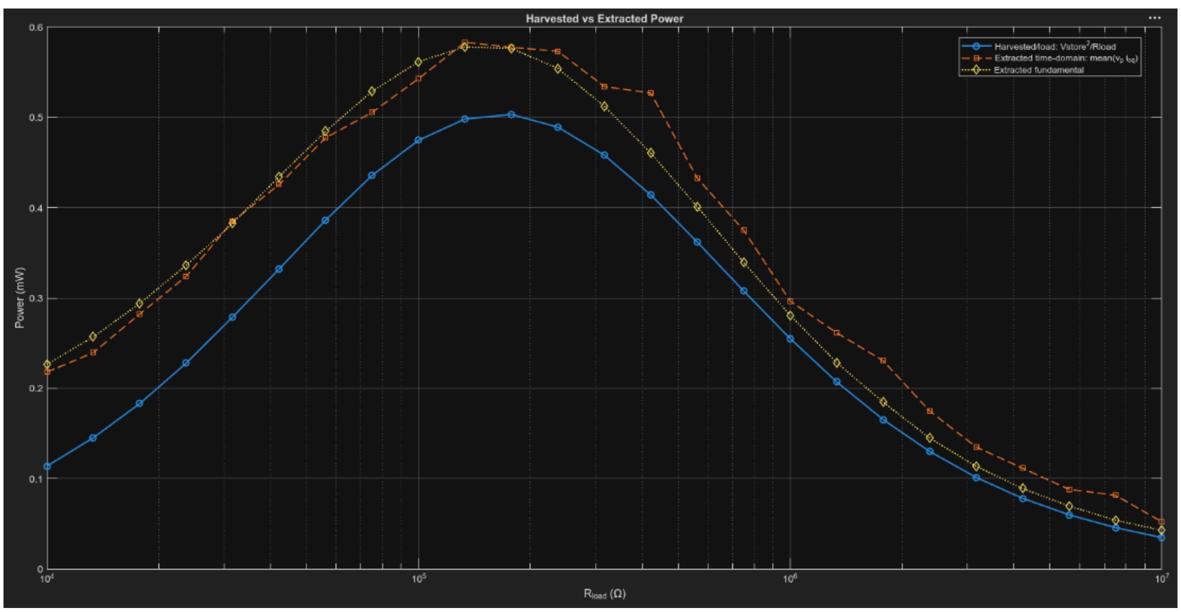


Figure 2.5: Representative SEH R_{load} sweep showing simulated harvested power and the analytical/fundamental power comparisons. TODO: change graph with added paper eqns waveform

2.5 Parallel SSHI (P-SSHI)

2.5.1 Ideal P-SSHI switching model

P-SSHI adds a switched inductor directly in parallel with the piezoelectric element. The switch is triggered at the displacement extremum, corresponding to the equivalent-current zero crossing in the ideal harmonic model. During the brief switching interval, L_i and C_p form an LC resonant loop and invert the piezoelectric voltage before the next half-cycle.

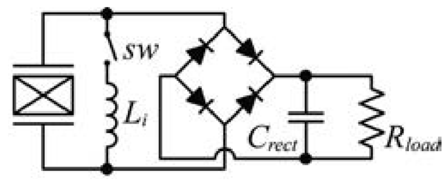


Figure 2.6: P-SSHI interface circuit topology. Figure taken from [1, Fig. 4].

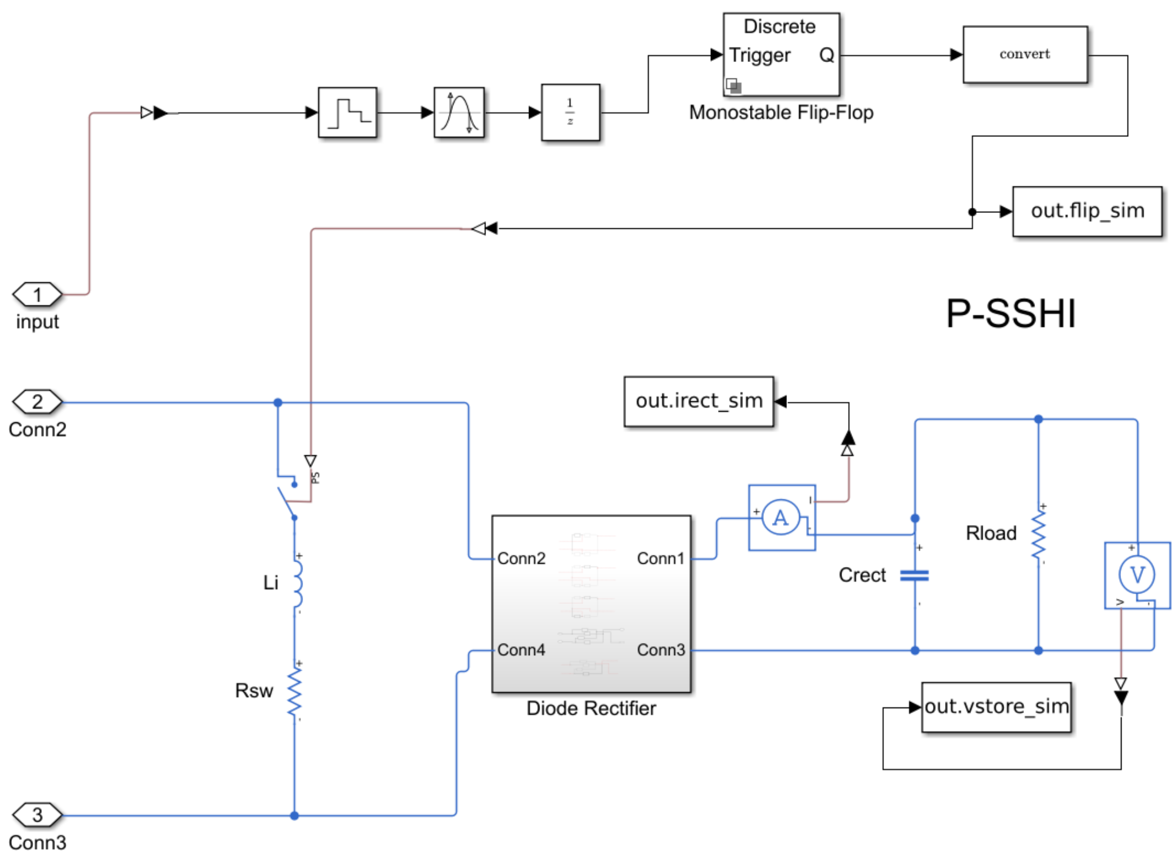


Figure 2.7: Internals of Simscape subsystem on the right from Figure 2.2 for P-SSHI interface circuit. Contains equivalent-current zero cross detector triggered for fixed duration T_{sw} (ideal non-circuit implementation) with ideal switch and inductor L_i/ resistor R_{sw}, followed by rectifier and smoothing capacitor C_{rect} with R_{load}.

The half-LC switching interval is

$$T_{sw} = \pi\sqrt{L_i C_p}, \tag{2.42}$$

which gives approximately 127 µs for $L_i = 47$ mH and the benchmark C_p. This is much shorter than the approximately 23.8 ms mechanical period at 42 Hz, so the analytical model treats the inversion as effectively instantaneous. TODO: Add Li, Cp switch graph

The voltage inversion factor used in the paper is related to the switching-loop quality factor by

$$\gamma = -\exp\left(-\frac{\pi}{2Q}\right). \tag{2.43}$$

where the quality factor Q (r is the resistance in the C_p-L_i-r loop including R_{sw}, and L_i/C_p series resistance) is:

$$Q = \frac{1}{r}\sqrt{\frac{L_i}{C_p}}. \tag{2.44}$$

The benchmark simulations used $\gamma = -0.7$.

2.5.2 P-SSHI waveform and impedance equations

For the same sinusoidal i_{eq} assumption, the P-SSHI piezoelectric voltage is

$$v_p(t) = \begin{cases} V_{oc}[1 - \cos(\omega t)] - \gamma V_{rect}, & 0^+ \le \omega t < \theta, \\ V_{rect}, & \theta \le \omega t \le \pi^-, \\ \gamma V_{rect} - V_{oc}[1 + \cos(\omega t)], & \pi^+ \le \omega t < \pi + \theta, \\ -V_{rect}, & \pi + \theta \le \omega t \le 2\pi^-. \end{cases} \tag{2.45}$$

The blocked angle now satisfies

$$\cos\theta = 1 - (1 + \gamma)\widetilde{V}_{rect}. \tag{2.46}$$

TODO: ref eqns/code

The resulting analytical waveform was overlaid with the simulated v_p. This comparison is more restrictive than comparing only power because it checks the rectifier-blocked interval, clamped interval, inversion event, and phase relative to i_{eq}.

Insert exported simulation figure

Export the final one-cycle P-SSHI plot showing simulated v_p, paper v_p, i_{eq}, fitted $i_{eq,F}$, V_{rect}, and the ideal switching pulse.

Figure 2.8: One-cycle P-SSHI waveform comparison between the Simscape result and Eq. (2.45).

The corresponding electrical equivalent impedance is

$$Z_{\text{elec,PSSHI}} = \frac{1}{\pi\omega C_p}\left\{(1-\cos\theta)\left[\frac{4}{1+\gamma} - 1 + \cos\theta\right] + j(\sin\theta\cos\theta - \theta)\right\}. \tag{2.47}$$

The P-SSHI impedance components used in the power calculation are

$$R_d = \frac{1}{\pi\omega C_p}\left\{2\widetilde{V}_F[2-\widetilde{V}_{rect}(1+\gamma)] + \widetilde{V}_{rect}^2(1-\gamma^2)\right\}, \tag{2.48}$$

$$R_h = \frac{2}{\pi\omega C_p}(\widetilde{V}_{rect} - \widetilde{V}_F)[2-\widetilde{V}_{rect}(1+\gamma)], \tag{2.49}$$

$$X_E = \frac{1}{\pi\omega C_p}(\sin\theta\cos\theta - \theta). \tag{2.50}$$

These values were then substituted into Eqs. (2.39) and (2.40) without changing the power equations themselves.

For a representative $1\,\mathrm{M\Omega}$ validation run, the simulated and analytical equivalent impedances were approximately

$$Z_{\text{elec,sim}} \approx (400 - j52.8)\,\mathrm{k\Omega}, \tag{2.51}$$

$$Z_{\text{elec,paper}} \approx (400 - j49.5)\,\mathrm{k\Omega}. \tag{2.52}$$

2.5.3 P-SSHI power and load sweep

The power comparison at the same operating point was

Table 2.3: Representative P-SSHI power comparison at $R_{load} = 1\,\text{M}\Omega$.

Quantity	Power
Simscape harvested/load power	2.09 mW
Eq. 42 harvested power	2.10 mW
Time-domain extracted power	3.83 mW
Fundamental extracted power	3.83 mW
Eq. 43 extracted power	3.71 mW
Predicted interface dissipation, $P_\Delta - P_h$	approximately 1.62 mW

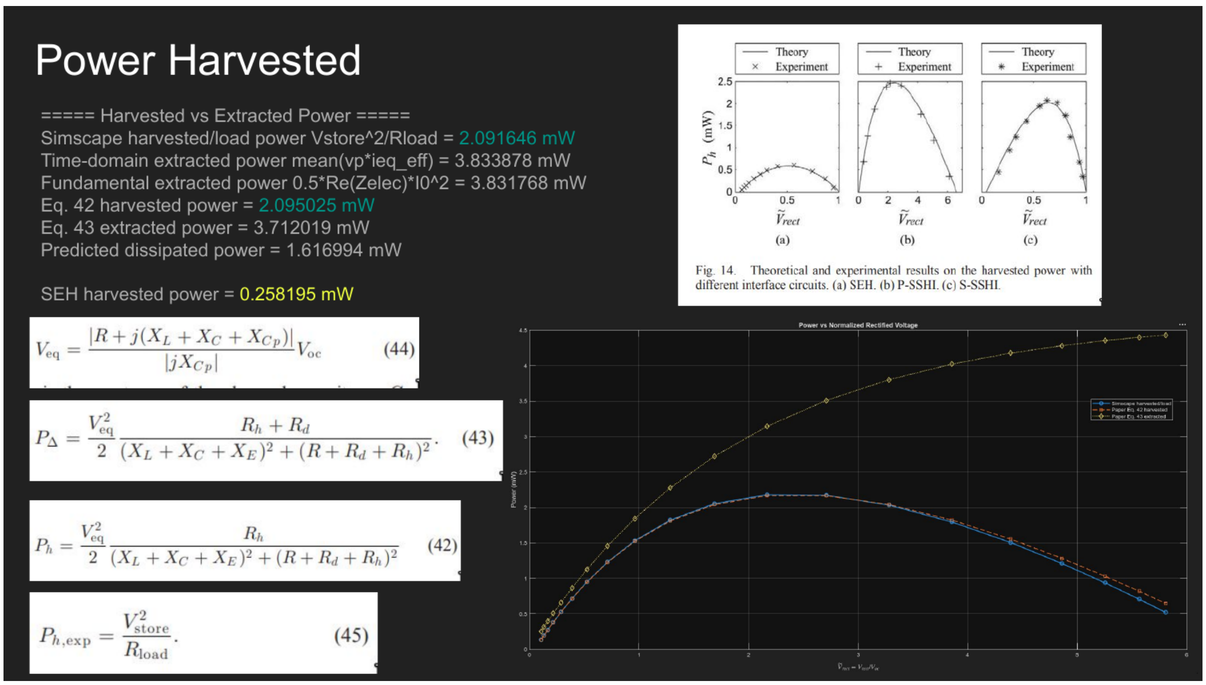


Figure 2.9: P-SSHI harvested and extracted power comparison. The simulated load-power curve closely follows Eq. 42, while the extracted fundamental follows the impedance-based extracted-power calculation. Liang and Liao's Fig. 14 is shown in the same development figure for comparison [1].

The same load-sweep code used for SEH was run with the P-SSHI variant active. The simulated harvested power rises as the output operating point becomes more favorable, reaches a maximum, and then falls as the increasingly high load draws too little useful dc current.

In contrast, the total extracted power can continue increasing over part of this range because a larger portion of the extracted energy is not necessarily delivered to the load. This distinction is exactly why both P_h and P_Δ were retained throughout the analysis.

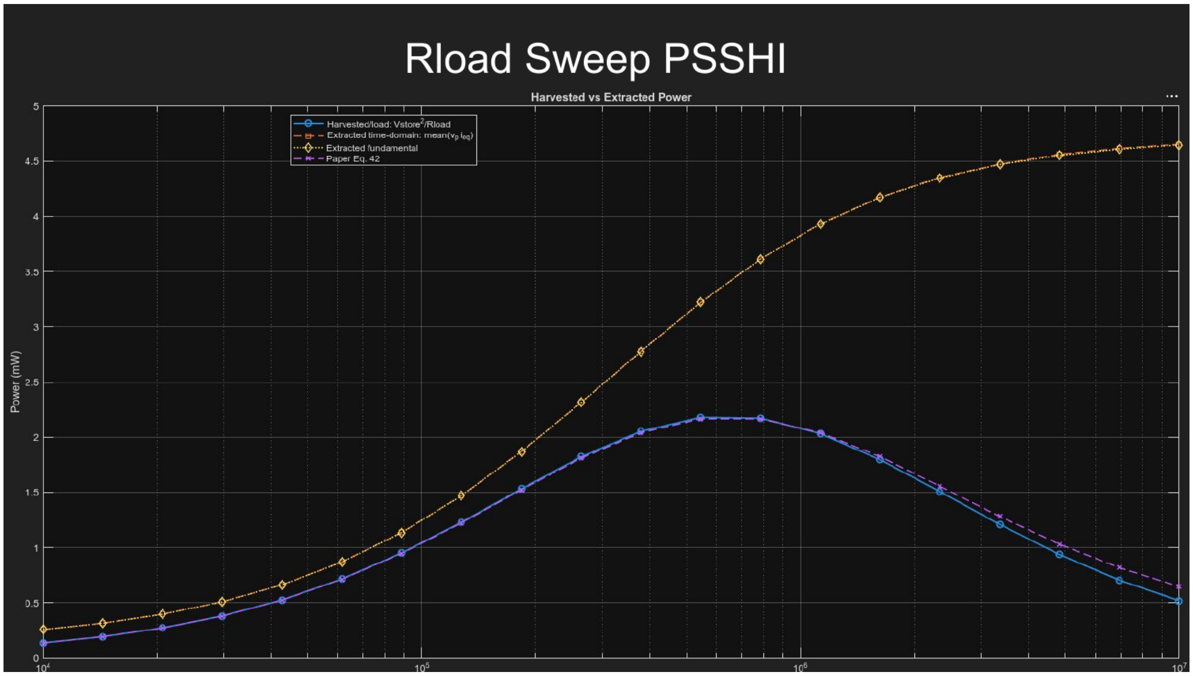


Figure 2.10: P-SSHI R_{load} sweep. The simulated harvested-power curve and Eq. 42 are close over the sweep, while the time-domain and fundamental extracted-power curves show the larger total energy extracted from the electromechanical system.

The final corrected P-SSHI sweep gave

$$R_{load,opt} \approx 546\,\mathrm{k\Omega}, \qquad P_{load,max} \approx 2.18\,\mathrm{mW}. \tag{2.53}$$

2.6 Series SSHI (S-SSHI)

2.6.1 Ideal S-SSHI switching model

S-SSHI places the switched inductor in series with the rectifier path rather than directly across the piezoelectric element. The same short inversion-time condition in Eq. (2.42) applies, but the resulting ideal voltage waveform differs from P-SSHI.

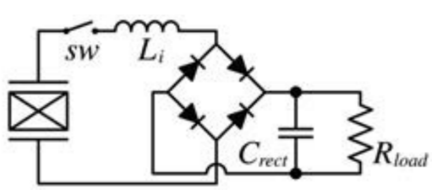


Figure 2.11: S-SSHI interface circuit topology. Figure taken from [1, Fig. 4].

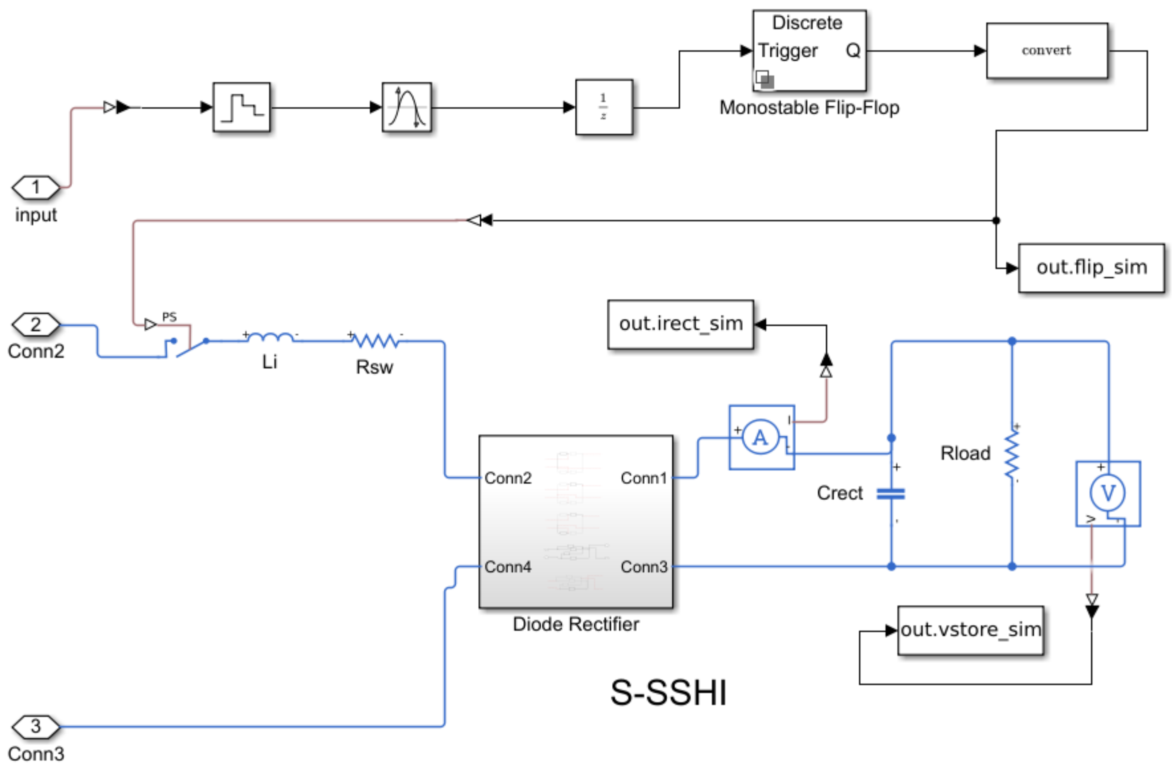


Figure 2.12: Internals of Simscape subsystem on the right from Figure 2.2 for S-SSHI interface circuit. Contains equivalent-current zero cross detector triggered for fixed duration T_{sw} (ideal non-circuit implementation) with ideal switch and inductor L_i/ resistor R_{sw}, followed by rectifier and smoothing capacitor C_{rect} with R_{load}.

2.6.2 S-SSHI waveform and impedance equations

Liang and Liao give

$$v_p(t) = \begin{cases} -V_{oc}\cos(\omega t) + \dfrac{1-\gamma}{1+\gamma}(V_{oc} - V_{rect}), & 0^+ \le \omega t \le \pi^-, \\ -V_{oc}\cos(\omega t) - \dfrac{1-\gamma}{1+\gamma}(V_{oc} - V_{rect}), & \pi^+ \le \omega t \le 2\pi^-. \end{cases} \tag{2.54}$$

Unlike SEH and P-SSHI, S-SSHI does not use the same rectifier blocked angle θ in the analytical waveform. Its equivalent impedance is

$$Z_{\text{elec,SSSHI}} = \frac{1}{\omega C_p}\left[\frac{4}{\pi}\frac{1-\gamma}{1+\gamma}(1-\widetilde{V}_{rect}) - j\right]. \tag{2.55}$$

The impedance components used in Eqs. (2.39) and (2.40) are

$$R_d = \frac{4}{\pi\omega C_p}\frac{1-\gamma}{1+\gamma}(1-\widetilde{V}_{rect}+\widetilde{V}_F)(1-\widetilde{V}_{rect}), \tag{2.56}$$

$$R_h = \frac{4}{\pi\omega C_p}\frac{1-\gamma}{1+\gamma}(\widetilde{V}_{rect}-\widetilde{V}_F)(1-\widetilde{V}_{rect}), \tag{2.57}$$

$$X_E = -\frac{1}{\omega C_p}. \tag{2.58}$$

TODO, anywhere screen prnt screenshot

As with SEH and P-SSHI, this paper waveform was compared against the actual Simscape $v_p(t)$ rather than being used to generate the simulated electrical response.

Insert exported simulation figure

Export the final one-cycle S-SSHI plot showing the simulated v_p, the Eq. (2.54) waveform, i_{eq}, and the synchronized switching event.

Figure 2.13: One-cycle S-SSHI waveform comparison between Simscape and the analytical waveform.

2.6.3 S-SSHI power and load sweep

The S-SSHI load sweep used exactly the same simulation and averaging procedure as SEH and P-SSHI. Its maximum occurred at a much smaller load resistance than the P-SSHI maximum:

$$R_{load,opt} \approx 61.6\,\mathrm{k\Omega}, \qquad P_{load,max} \approx 1.75\,\mathrm{mW}. \tag{2.59}$$

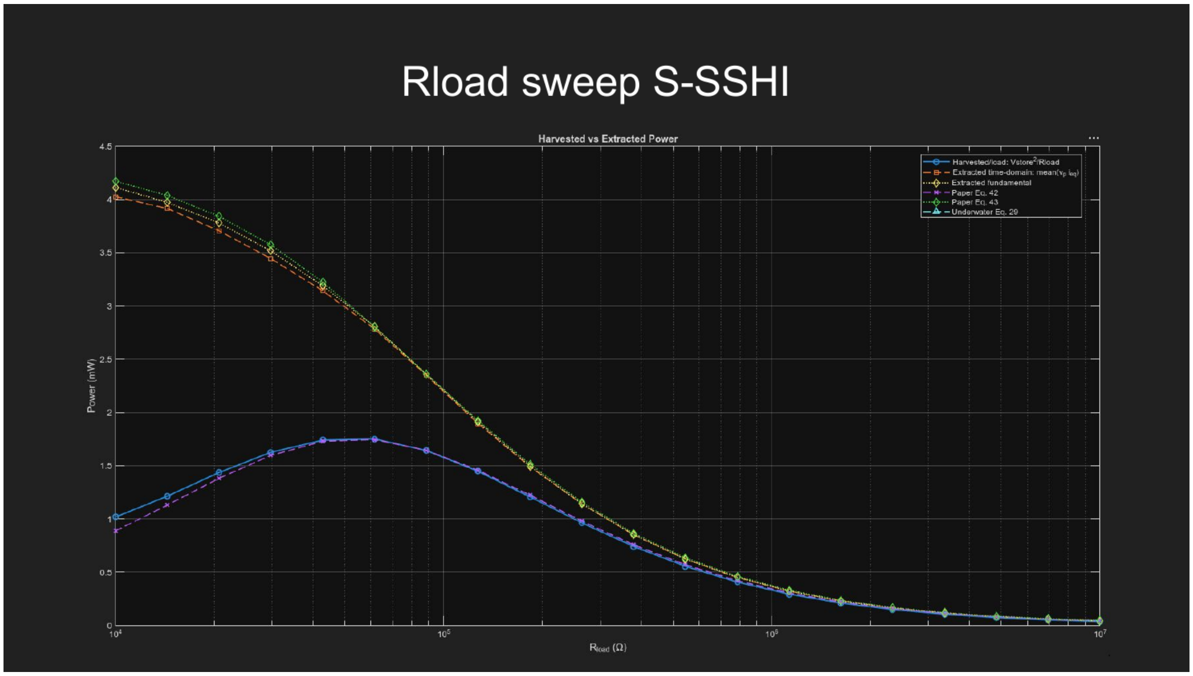


Figure 2.14: S-SSHI R_{load} sweep. The useful harvested-power maximum is lower than P-SSHI but substantially higher than SEH for the benchmark source.

2.7 Direct Comparison with Liang and Liao

2.7.1 Waveform comparison

The first validation level was waveform shape. The simulated SEH waveform reproduced the blocked interval followed by the $\pm V_{rect}$ clamping in Eq. (2.29). P-SSHI reproduced the abrupt inversion followed by the blocked and clamped intervals in Eq. (2.45). S-SSHI reproduced the alternating offset in Eq. (2.54).

In all three cases, the comparison was made after extracting I_0, phase, and V_{rect} from the simulation itself. The code first used the fitted I_0 to obtain V_{oc}, then used the simulated V_{rect} to calculate $\widetilde{V}_{rect}$ and θ, and finally evaluated each V_p equation point by point.

P-SSHI and S-SSHI both use the same SSHI inversion factor gamma. SEH is the gamma = 1 special case of the P-SSHI formulas.

```
TODO: label/refer code
Qsshi = sqrt(Li/Cp) / Rloop_total;
gamma_sshi = -exp(-pi/(2*Qsshi));
if Type == "SEH"
    gamma = 1;
```

```
elseif Type == "PSSHI" || Type == "SSSHI"
    gamma = gamma_sshi;

Voc_sim = I0 / (w*Cp);
Vrect_sim = Vstore_avg + VF_bridge;
Vtilde_sim = Vrect_sim / Voc_sim;
theta_paper = acos(1 - (1 + gamma)*Vtilde_sim);

phi = mod(w*t_c + ieq_phase, 2*pi);
vp_paper = nan(size(phi));

if Type == "SSSHI"
    offset = ((1 - gamma)/(1 + gamma)) * (Voc_sim - Vrect_sim);

    for kk = 1:length(phi)
        p = phi(kk);
        if p >= 0 && p < pi
            vp_paper(kk) = -Voc_sim*cos(p) + offset;
        else
            vp_paper(kk) = -Voc_sim*cos(p) - offset;
        end
    end

elseif Type == "SEH" || Type == "PSSHI"

    for kk = 1:length(phi)
        p = phi(kk);

        % General SEH/P-SSHI Eq. 21. SEH is the special case gamma = 1.
        if p >= 0 && p < theta_paper
            vp_paper(kk) = Voc_sim * (1 - cos(p)) - gamma * Vrect_sim;
        elseif p >= theta_paper && p < pi
            vp_paper(kk) = Vrect_sim;
        elseif p >= pi && p < pi + theta_paper
            vp_paper(kk) = gamma * Vrect_sim - Voc_sim * (1 + cos(p));
        else
            vp_paper(kk) = -Vrect_sim;
        end
```

```
    end
end
```

2.7.2 Power comparison using Eqs. 42 and 43

The third validation level was power. The following table collects the representative SEH and P-SSHI comparisons used during development:

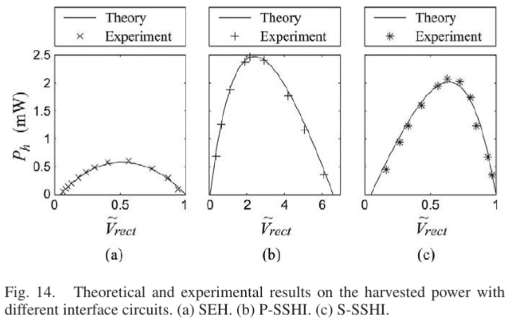
Table 2.4: Comparison of simulated power and Liang–Liao analytical power at representative $1\,\mathrm{M\Omega}$ operating points.

Quantity	SEH	P-SSHI
Simulated harvested/load power	0.255 mW	2.09 mW
Eq. 42 harvested power	0.258 mW	2.10 mW
Time-domain extracted $\langle v_p i_{eq} \rangle$	0.321 mW	3.83 mW
Fundamental extracted power	0.282 mW	3.83 mW
Eq. 43 extracted power	0.274 mW	3.71 mW

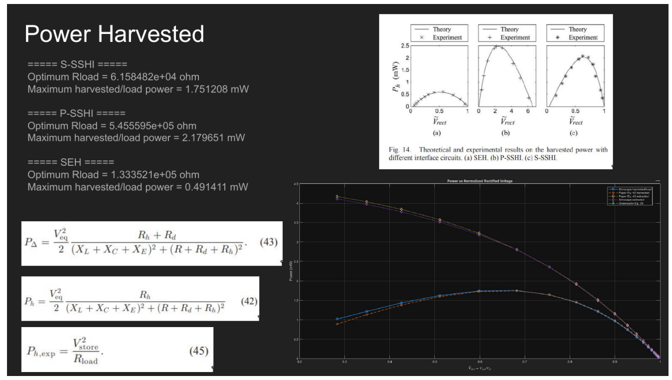
This comparison is important because harvested and extracted power are not the same quantity. V_{store}^2/R_{load} measures useful dc power delivered to the load, while $\langle v_p i_{eq} \rangle$ measures the total real electrical power removed from the equivalent source. Eq. 42 predicts the harvested branch R_h, whereas Eq. 43 includes both R_h and the dissipative branch R_d.

2.7.3 Comparison with Fig. 14

Liang and Liao's Fig. 14 compares theoretical and experimental harvested power for SEH, P-SSHI, and S-SSHI. Their P-SSHI case produced the greatest harvestable power, followed by S-SSHI and then SEH [1]. The Simscape load sweeps reproduced the same ordering.



(a) Published theoretical/experimental curves.



(b) Simscape comparison and final simulated maxima.

Figure 2.15: Comparison with Liang and Liao Fig. 14 [1]. The exact horizontal variables differ between the development plots, but the simulated interfaces reproduce the same load-dependent maxima and the same ordering of P-SSHI, S-SSHI, and SEH.

The final corrected Simscape load-sweep maxima were

Table 2.5: Final vibration-benchmark maxima from the Simscape load sweeps.

Interface	Optimum R_{load}	Maximum load power	Relative to SEH
SEH	133 kΩ	0.49 mW	1.0×
S-SSHI	61.6 kΩ	1.75 mW	3.6×
P-SSHI	546 kΩ	2.18 mW	4.4×

Thus, ideal P-SSHI produced the largest harvested power for the vibration benchmark. More importantly for the remainder of the work, the close waveform, impedance, Eq. 42/Eq. 43, and load-sweep agreement showed that the Simscape model was sufficiently consistent with the published analytical framework to use as the baseline for replacing the ideal switching action with real switching circuitry and, later, for changing the source to the underwater equivalent model.

Chapter 3

Self-Powered and Alternative SSHI Switching

3.1 Reference Self-Powered P-SSHI Circuit

Zhang et al. proposed an efficient self-powered P-SSHI interface in which passive peak-detection branches trigger the inductor path without an externally powered timing source [2]. The published circuit uses separate positive and negative detection paths so that the piezoelectric capacitance and an inductor form the inversion loop after the detected voltage peak.

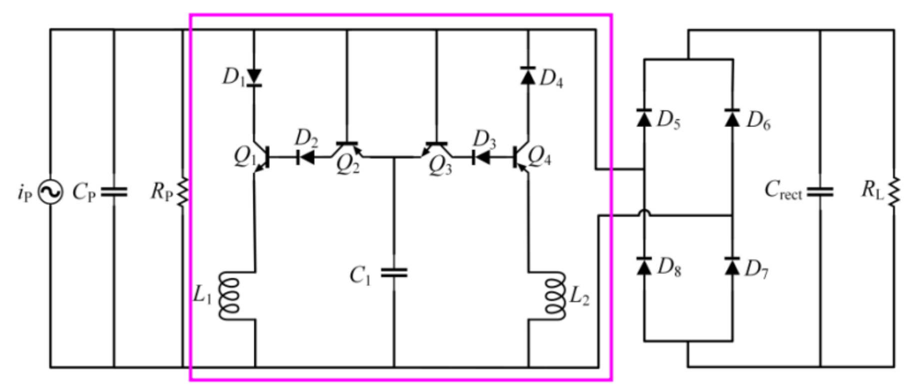


Figure 3.1: ESP-PSSHI circuit used as the self-powered switching implementation. The i_p, C_p and R_p is a simplified piezo model of the $V_{eq} + RLC$ and C_p circuit model in Figure 2.1 and the ideal switch is replaced with the outlined area. Figure taken from [2, Fig. 6]

The Enhanced Self-Powered Parallel Synchronous Switch Harvesting on Inductor (ESP-PSSHI) circuit eliminates the need for external digital controllers, active sensors, or external power supplies by using a passive peak-detector and envelope-tracking configuration to trigger voltage inversion.

3.1.1 Circuit Phases and Operating Principles

The operation of the ESP-PSSHI circuit occurs in three distinct phases over each half-cycle of the piezoelectric excitation:

1. **Peak Tracking Phase (v_p Rising):** As the piezoelectric element experiences mechanical strain, its internal equivalent current source i_p charges the clamped capacitance C_p, causing the piezoelectric voltage v_p to increase. During this period, diode D_2 becomes forward-biased, charging the envelope-tracking capacitor C_1 such that its voltage tracks v_p minus a diode drop. The switching transistor Q_1 remains off since its base-emitter junction is not sufficiently forward-biased.

2. **Envelope Tracking and Peak Detection (v_p Falling):** When the mechanical displacement reaches an extremum, i_p reverses direction and v_p begins to decrease. However, capacitor C_1 holds the peak voltage $v_{p,\max}$. Once the potential difference between v_{C1} and v_p exceeds the conduction threshold of the BJT-diode pair, current flows into the base of Q_1, driving it into saturation and triggering the switching branch.

3. **Resonant LC Inversion Phase:** The conduction of Q_1 creates a closed resonant path formed by C_p, diode D_1, Q_1, and inductor L_1. Current rapidly flows through L_1, inverting the charge across C_p. Diode D_1 prevents reverse current flow after half an LC cycle, naturally extinguishing the branch current once the inductor current returns to zero. An identical complementary network (D_3, Q_3, Q_4, D_4, L_2) handles the negative half-cycle.

3.1.2 Mathematical Formulation

The switching action is initiated when the voltage difference between the peak-hold capacitor C_1 and the decreasing piezoelectric voltage v_p exceeds the required semiconductor threshold voltage:

$$v_{C1} - v_p \geq V_{BE} + V_D \tag{3.1}$$

where V_{BE} is the base-emitter turn-on voltage of transistor Q_1 and V_D is the forward voltage drop of the series diode.

During the inversion phase, the closed loop formed by C_p, L_1, and total loop resistance R_{total} behaves as a second-order series RLC circuit. The duration of the resonant switching pulse corresponds to half the natural LC period:

$$T_{sw} = \pi\sqrt{L_1 C_p} \tag{3.2}$$

The attenuation of the piezoelectric voltage during inversion is dictated by the quality factor Q of the switching loop:

$$Q = \frac{1}{R_{\text{total}}}\sqrt{\frac{L_1}{C_p}} \tag{3.3}$$

where R_{total} accounts for the inductor equivalent series resistance (ESR), transistor on-resistance, diode resistance, and wiring traces.

The voltage inversion factor γ, defined as the ratio of the piezoelectric voltage immediately after switching to the voltage immediately before switching, is expressed as:

$$\gamma = -e^{-\frac{\pi}{2Q}} \tag{3.4}$$

Consequently, the piezoelectric voltage $v_p(t^+)$ immediately following the inversion process relates to its pre-switch peak value $v_p(t^-)$ via:

$$v_p(t^+) = \gamma \cdot v_p(t^-) \tag{3.5}$$

3.2 Measured Trigger Delay and Inversion Duration

The important distinction in this simulation was between *how long* the switch remained on and *when* it began to conduct. From Eq. (2.42), using $L_i = 47\,\text{mH}$ and the benchmark C_p, the expected ideal half-LC interval was

$$T_{sw} = \pi\sqrt{L_i C_p} = 126.853\,\mu\text{s}. \tag{3.6}$$

The timing measured from the final steady-state cycle was:

Table 3.1: Measured switching timing in the benchmark P-SSHI simulation.

Switch implementation	Closed duration	Delay from i_{eq} zero crossing
Ideal/control switch	124.916 µs	40.525 µs
BJT self-powered detector	133.849 µs	950.103 µs

The BJT duration was close to the expected LC interval, but the branch began nearly 1 ms after the desired event. This explains why changing only the inductor or the on-time does not fully recover the ideal P-SSHI result: the inversion must also begin at the correct phase.

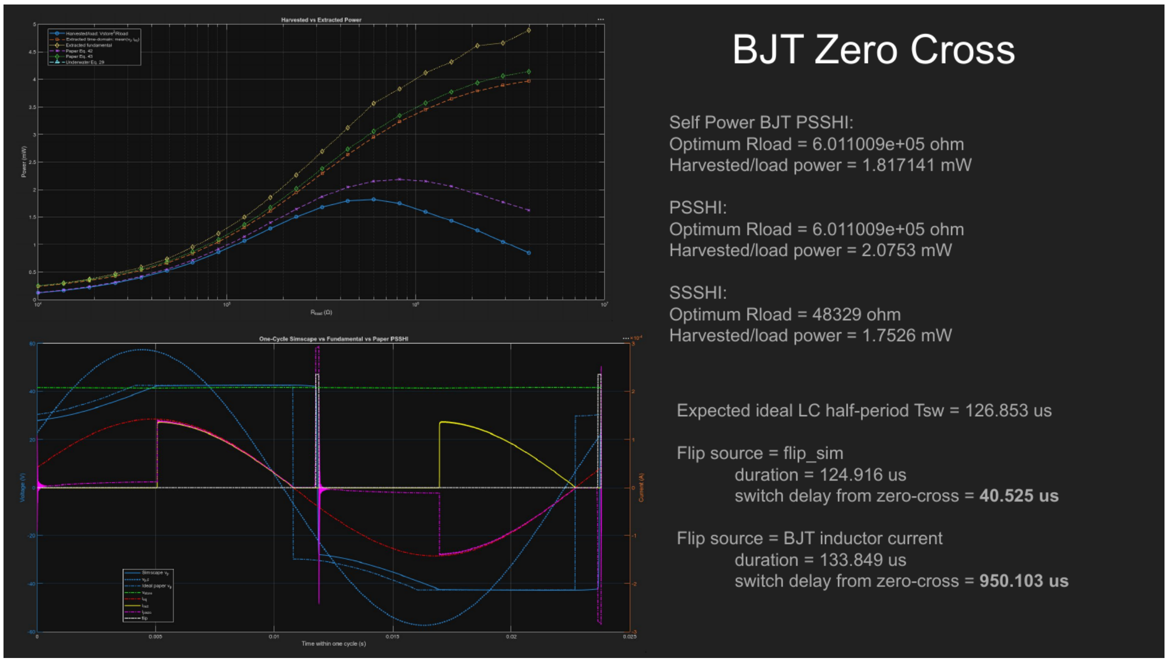


Figure 3.2: Measured ideal-switch and BJT switching timing and the corresponding load-sweep results. The plotted timing was measured directly from the logged control signal or BJT inductor current.

At the tested load sweep, the self-powered BJT P-SSHI maximum was about 1.82 mW, compared with about 2.08 mW for the ideal-switch P-SSHI configuration at the same sweep settings. The BJT circuit therefore retained most of the SSHI benefit, but its delay was a measurable source of loss.

3.3 Passive Detector Component Sweep

The passive detector was also swept with real transistor and diode models at underwater source levels. At 230 dB, the baseline BAT54 and TIP31C/TIP32C implementation produced about 35 µW, while the best tested combination using BSS4130A, BSS5130A, and MBR0520L produced about 56 µW. At 250 dB, the corresponding values were about 20.5 mW and 21.4 mW. These results show that device selection matters, but the importance of forward drop is specific to the low-voltage underwater case; it is not the main conclusion of the overall work.

3.4 Two Active Zero-Cross / Peak-Detection Variants

Two additional active timing circuits were implemented in LTspice to test whether a powered detector could trigger the inversion more accurately. Their schematics and power curves are shown in Figures 3.3 and 3.4.

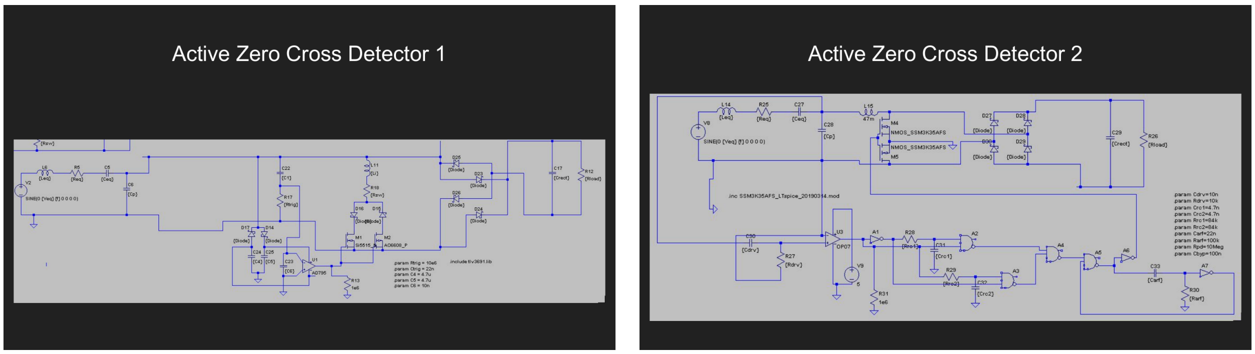


Figure 3.3: Two active zero-cross/peak-detection variants tested in LTspice. They are exploratory implementations; the slide record does not identify either schematic as a direct reproduction of one specific published circuit.

Figure 3.4: Power comparison of the previous passive P-SSHI detector, SEH, active detector 1, and active detector 2. The passive P-SSHI remained the best of the tested timing circuits; active detector 2 approached it more closely than active detector 1.

From the plotted sweep, the best point of the passive detector was approximately 35 µW, active detector 2 approximately 33 µW, active detector 1 approximately 24 µW, and

SEH approximately 22 µW. These values are read from the simulation plot and are therefore reported only to engineering precision.

The likely limitation was timing rather than simply lack of active circuitry. The tested active versions detected a voltage derivative or voltage peak, whereas the desired inversion start in the simulated P-SSHI waveform was tied to the phase of the equivalent current. Related literature confirms that switching-time generation is a central SSHI design problem. Zouari et al. compared op-amp/MOSFET and bipolar self-powered switching techniques and reported much higher simulated power from the bipolar approach after loss reduction [7]. Wu et al. instead integrated SSHI with active diodes to control the inversion interval automatically; their measured circuit harvested 2.1 times the power of a full bridge with 85% conversion efficiency [8]. These circuits are useful references for a deeper next-stage timing study, rather than evidence that either active schematic tested here was already optimal.

Chapter 4

Underwater Acoustic Equivalent Source and Frequency Effects

4.1 Equivalent Circuit Used in the Simulations

Lutz Lampe developed frequency-domain acoustic–mechanical–electrical equivalent circuits for the underwater transducer used in this work [3, 4]. The model converts incident acoustic pressure into an electrical Thevenin source with a series equivalent branch and piezoelectric capacitance. The simulation only requires the resulting terminal-equivalent source; the internal derivation is not itself an optimization variable.

Two forms of the equivalent circuit were used during the work. The older form and the newer clamped-capacitance form have very different internal V_{eq}, R_{eq}, L_{eq}, C_{eq}, and C_p values, but when transformed consistently they present essentially the same source at the piezoelectric terminals. Consequently, the choice between the two forms does not change the circuit-design conclusions.

For example, at 42 Hz and 230 dB, the old representation gives approximately $R_{eq} = 520\,\mathrm{M}\Omega$, $L_{eq} = 68\,\mathrm{H}$, $C_{eq} = 579\,\mathrm{nF}$, and $C_p = 336.5\,\mathrm{nF}$, whereas the new representation gives approximately $R_{eq} = 0.62\,\Omega$, $L_{eq} = 29.5\,\mathrm{nH}$, $C_{eq} = 211\,\mathrm{nF}$, and $C_p = 125\,\mathrm{nF}$. Despite this, both produce approximately $1.73\,\mathrm{V}$ open circuit at the piezoelectric port.

4.2 Numerical Check: Old and New Forms Give the Same Result

The terminal-equivalence claim was checked directly by running the same rectifier and switching simulations with both models. Representative results are in Table 4.1.

Table 4.1: Representative old/new underwater-model power comparison.

Case	Old model	New model	Difference
SEH, 230 dB	18.68 μW	18.66 μW	negligible
SEH, 250 dB	3.59 mW	3.58 mW	negligible
BJT P-SSHI, 230 dB	29.55 μW	29.50 μW	negligible
BJT P-SSHI, 250 dB	21.16 mW	21.13 mW	negligible

At 250 dB, both forms also gave $|V_{oc}| \approx 17.28\,\mathrm{V}$ and the same terminal output impedance in the comparison calculation. The remainder of the report therefore uses whichever representation is convenient for a particular simulation and focuses on the observable terminal parameters.

4.3 Source Level

For the representative $A = 2 \times 10^{-3}\,\mathrm{m}^2$ PZT element at 42 Hz, the terminal open-circuit amplitude scales with acoustic pressure. Three useful reference points are:

Table 4.2: Representative underwater open-circuit voltage levels at 42 Hz.

Acoustic level	$\lvert V_{oc}\rvert$
180 dB	5.46 mV
230 dB	1.73 V
250 dB	17.28 V

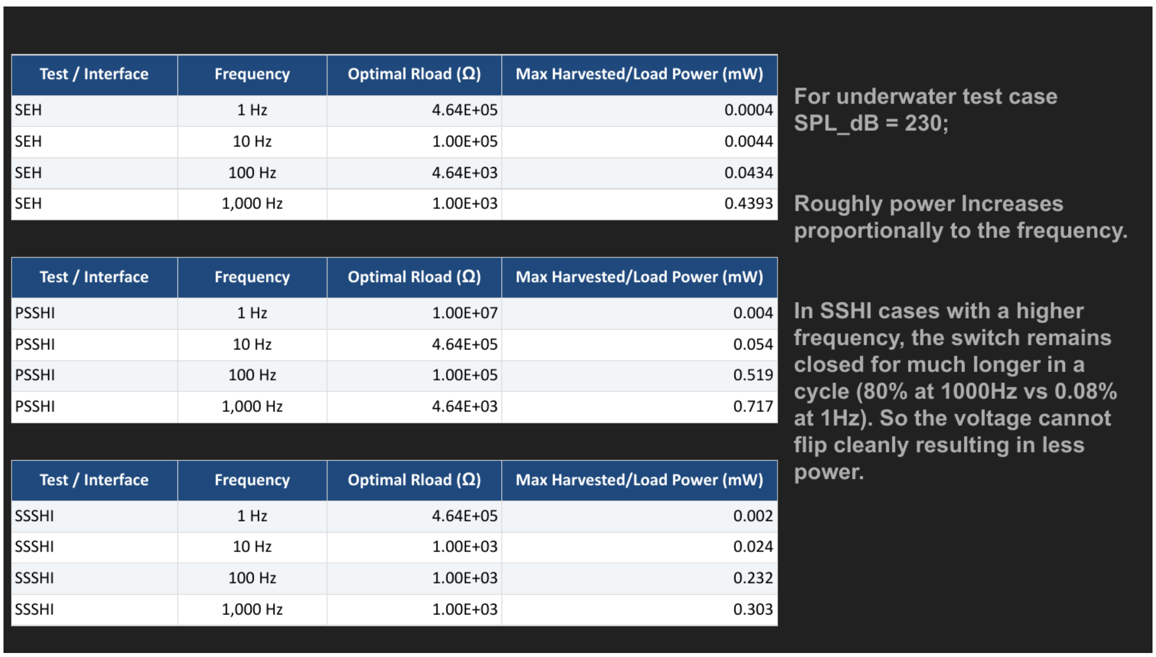
The 180 dB point is the regime in which semiconductor thresholds and leakage can dominate. The 230 dB point is useful for comparing practical switching and transformer matching because the open-circuit voltage is already high enough that a conventional bridge can conduct, while the source impedance remains strongly reactive.

4.4 Frequency Sweep and SSHI On-Time

A frequency sweep at 230 dB was performed while keeping $L_i = 47\,\mathrm{mH}$ fixed. The maximum load powers were:

Table 4.3: Underwater frequency sweep at 230 dB with fixed SSHI inductor.

Frequency	SEH	P-SSHI	S-SSHI
1 Hz	0.0004 mW	0.004 mW	0.002 mW
10 Hz	0.0044 mW	0.054 mW	0.024 mW
100 Hz	0.043 mW	0.519 mW	0.232 mW
1000 Hz	0.439 mW	0.717 mW	0.303 mW



Test / Interface	Frequency	Optimal Rload (Ω)	Max Harvested/Load Power (mW)
SEH	1 Hz	4.64E+05	0.0004
SEH	10 Hz	1.00E+05	0.0044
SEH	100 Hz	4.64E+03	0.0434
SEH	1,000 Hz	1.00E+03	0.4393

Test / Interface	Frequency	Optimal Rload (Ω)	Max Harvested/Load Power (mW)
PSSHI	1 Hz	1.00E+07	0.004
PSSHI	10 Hz	4.64E+05	0.054
PSSHI	100 Hz	1.00E+05	0.519
PSSHI	1,000 Hz	4.64E+03	0.717

Test / Interface	Frequency	Optimal Rload (Ω)	Max Harvested/Load Power (mW)
SSSHI	1 Hz	4.64E+05	0.002
SSSHI	10 Hz	1.00E+03	0.024
SSSHI	100 Hz	1.00E+03	0.232
SSSHI	1,000 Hz	1.00E+03	0.303

Figure 4.1: Frequency sweep and optimum-load results from the underwater Simscape model.

The raw power tends to increase with frequency, but the fixed inversion inductor becomes progressively less appropriate. With $L_i = 47\,\mathrm{mH}$, Eq. (2.42) gives approximately $395\,\mu\mathrm{s}$. That is only about 0.08% of a half-cycle at 1 Hz but about 80% of a half-cycle at 1000 Hz. In the SSHI simulations, the switch therefore remains closed for a much larger fraction of the mechanical cycle as frequency increases. The voltage can no longer complete a clean, short inversion, and the P-SSHI waveform becomes increasingly SEH-like.

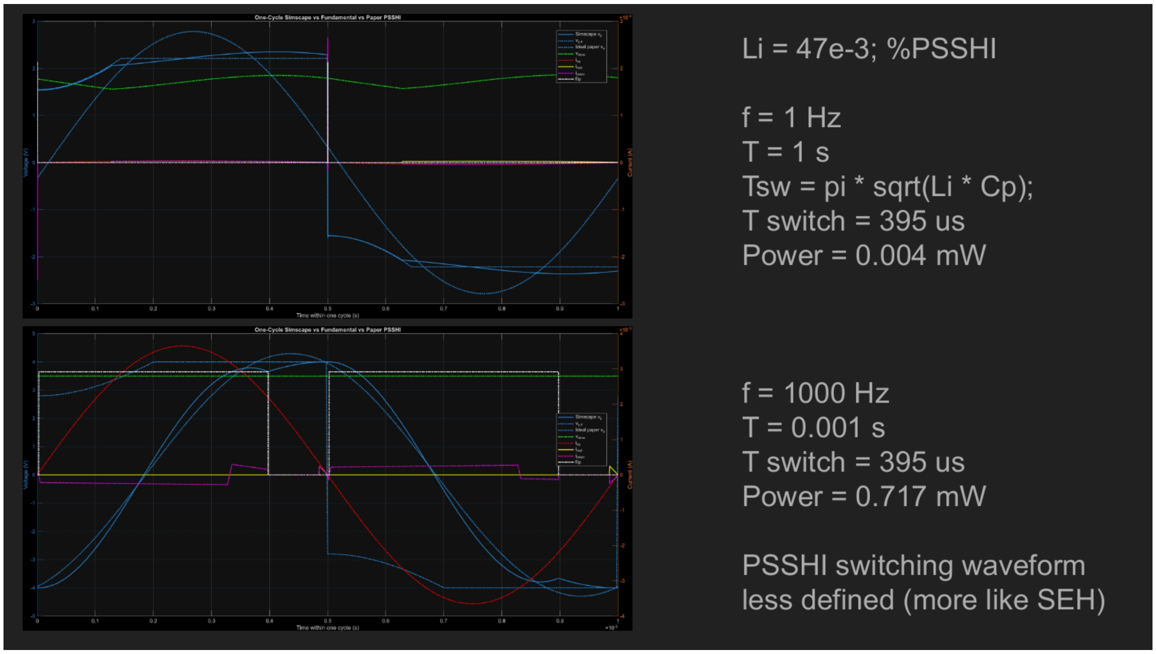


Figure 4.2: P-SSHI switching with fixed $L_i = 47\,\text{mH}$ at 1 Hz and 1000 Hz. The same absolute switching time occupies a much larger fraction of the high-frequency cycle.

Reducing the inversion inductance at 1000 Hz restored a short switching event. With $L_i \approx 188\,\mu\text{H}$, the simulated inversion time was about $25\,\mu\text{s}$ and the harvested power increased to about $5.9\,\text{mW}$.

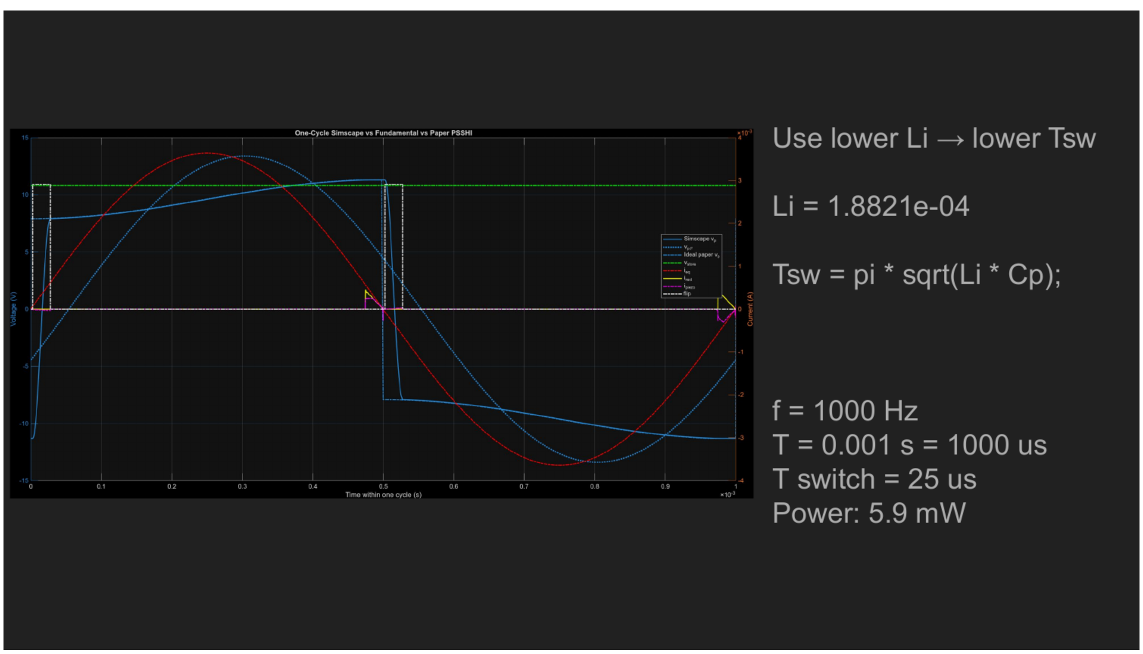


Figure 4.3: High-frequency P-SSHI result after reducing L_i so that the LC inversion occupies a small fraction of the 1000 Hz cycle.

A separate same-frequency sweep showed that L_i itself was not highly sensitive while the switching interval remained short enough: P-SSHI results for 1 mH, 10 mH, 100 mH, and 1 H were approximately 0.23, 0.22, 0.25, and 0.23 mW in that test. The important quantity is therefore the ratio of T_{sw} to the mechanical period, together with switching-loop loss, not simply a single "best" inductance independent of frequency.

Chapter 5

LTspice Transformer Matching and Real-Component Sweeps

5.1 Why the Simulation Moved to LTspice

The Simscape model was useful for reproducing the analytical paper and establishing an ideal-switch upper bound. After that validation, continuing to sweep ideal switches was no longer the main need. The circuit questions had become device-level questions: which diode leakage model is acceptable, how much transformer winding resistance can be tolerated, what primary inductance is required, how sensitive is the match to frequency, and what happens when actual catalog component models are inserted.

LTspice was therefore used because it allowed much faster parameter sweeps and direct use of manufacturer-style SPICE models for diodes, BJTs, MOSFETs, inductors, and coupled windings. Ideal switching remained only as a reference where useful; the optimization itself moved toward real components.

5.2 Reactive Matching Condition for the Transformer Primary

The underwater piezoelectric terminal is strongly capacitive. Consider the piezoelectric capacitance C_p in parallel with a transformer primary magnetizing inductance L_p. Their susceptances are

$$B_C = \omega C_p, \tag{5.1}$$
$$B_L = -\frac{1}{\omega L_p}. \tag{5.2}$$

At a chosen operating frequency, the first-order reactive match occurs when these cancel:

$$\omega C_p - \frac{1}{\omega L_p} = 0, \tag{5.3}$$

so

$$\boxed{L_p = \frac{1}{\omega^2 C_p}}. \tag{5.4}$$

This equation is a *frequency-specific shunt-resonance condition*. It is not by itself the general maximum-power-transfer theorem. It removes the dominant capacitive susceptance at one frequency so that a larger fraction of the source current can be transferred into the resistive load branch rather than circulating reactively through C_p.

For the old Lampe representation at 42 Hz, $C_p \approx 336.5\,\mathrm{nF}$, giving

$$L_p \approx 42.7\,\mathrm{H}. \tag{5.5}$$

This explains why the low-frequency transformer problem is difficult: tens of henries of primary inductance are required before turns ratio is even considered.

5.3 Turns Ratio and Reflected Load

In this report the physical transformer turns ratio is defined as

$$n = \frac{N_s}{N_p}. \tag{5.6}$$

For an ideal transformer,

$$\frac{V_s}{V_p} = n, \qquad \frac{I_s}{I_p} = \frac{1}{n}, \qquad L_s \approx n^2 L_p. \tag{5.7}$$

A secondary load Z_L is reflected to the primary as

$$Z_{L,\mathrm{ref}} = \left(\frac{N_p}{N_s}\right)^2 Z_L = \frac{Z_L}{n^2}. \tag{5.8}$$

Thus, L_p primarily controls the reactive match at the source frequency, while n and R_{load} determine the effective resistive loading seen at the primary. Some early MATLAB sweeps used a variable called `WindingRatio` with the reciprocal convention; the results in this report use the physical definition above to avoid ambiguity.

5.4 Relation to Previous Impedance-Matching Work

Transformer-based impedance matching is not new. Jabbar et al. proposed piezoelectric energy-harvester impedance matching using a step-down *piezoelectric* transformer and an inductor-less bidirectional half bridge [5]. Their motivation was closely related to the present result: a piezoelectric source has a high, capacitive impedance, and effective power transfer requires both resistive transformation and compensation of the reactive part. Their piezoelectric transformer also reduced the high harvester voltage and avoided the large magnetic component used in conventional approaches.

The magnetic-transformer version examined here is less common than SSHI for low-frequency harvesters for practical reasons. Equation (5.4) can require very large magnetizing inductance at tens of hertz. A transformer with that inductance is physically large, expensive, and usually has non-negligible winding resistance and core loss. In addition, the cancellation is narrowband: a small change in f, C_p, or L_p destroys the reactive match. P-SSHI uses a much smaller inductor only during a brief inversion interval and therefore does not require the magnetic component to remain reactively matched during the entire vibration cycle. The transformer can nevertheless be much better at a carefully tuned operating point, which is exactly what the LTspice results showed.

5.5 Direct Transformer versus P-SSHI Comparison at 230 dB

A direct LTspice comparison used the same underwater source at 42 Hz and 230 dB, corresponding to $|V_{oc}| \approx 1.73\,\mathrm{V}$. The transformer used $L_p = L_s \approx 42.68\,\mathrm{H}$ with $300\,\Omega$ series resistance in each winding, while the P-SSHI branch used a $15\,\mathrm{mH}$ inductor with about $3.39\,\Omega$ series resistance.

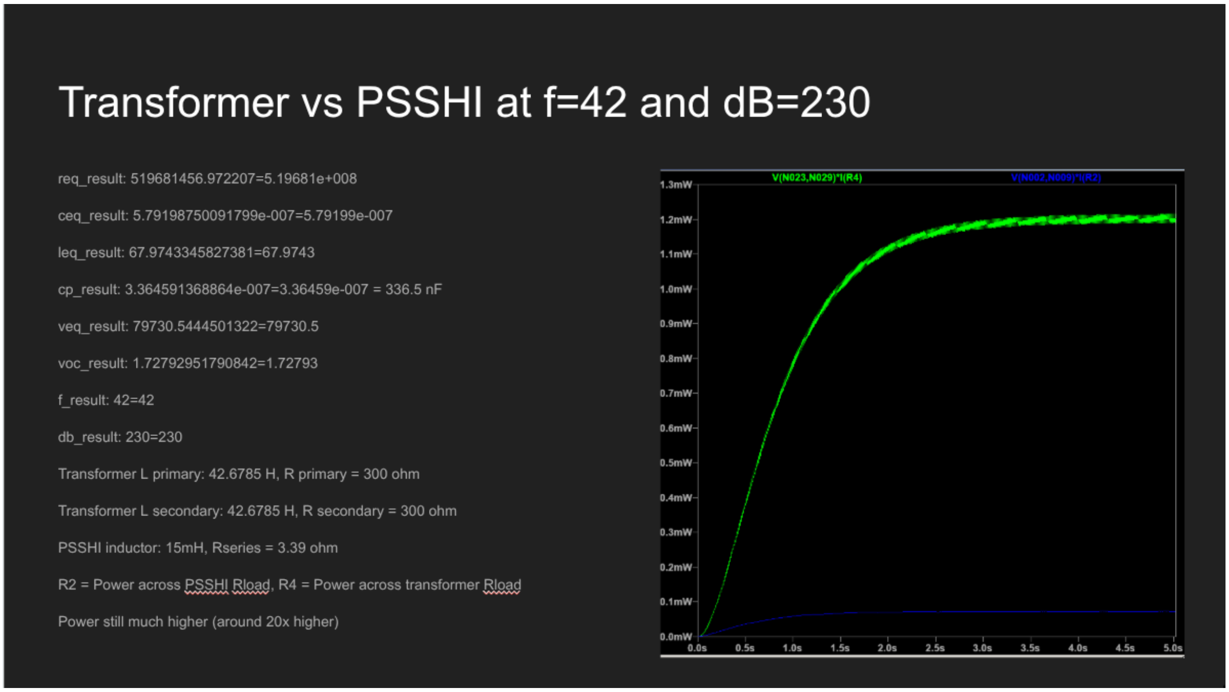


Figure 5.1: Direct LTspice comparison of transformer matching and P-SSHI at 42 Hz and 230 dB. The transformer load trace settles near 1.2 mW, while the compared P-SSHI trace is near 0.07 mW, an advantage on the order of 20 times at this tuned point.

This result is important because the transformer advantage did not appear only in the 5.46 mV case where diode forward voltage is a dominant issue. It also appeared at an open-circuit voltage of about 1.73 V. The advantage therefore came from impedance/reactive matching as well as voltage transformation.

5.6 Sensitivity to L_p and C_p

The same comparison also showed why the transformer result should not be interpreted as a universal replacement for SSHI. Small parameter changes caused large power changes.

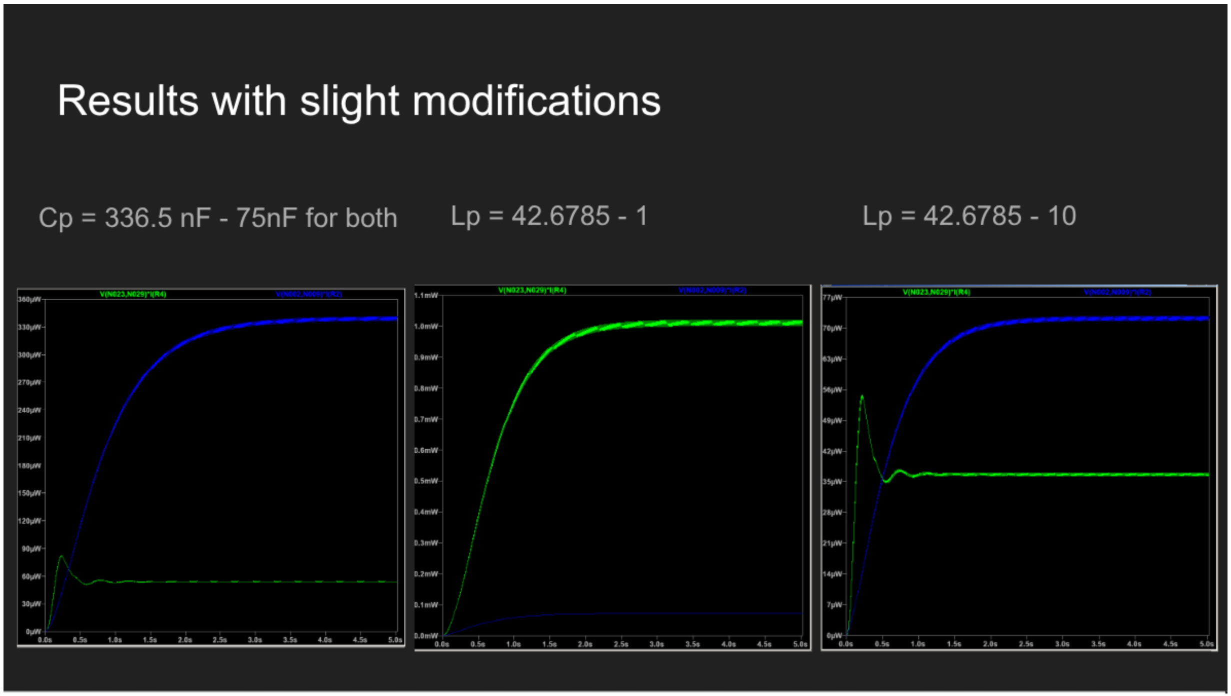


Figure 5.2: Sensitivity of the 230 dB transformer/P-SSHI comparison. Reducing the modeled C_p by 75 nF or shifting L_p away from its 42.68 H target can reduce or reverse the transformer's advantage.

Approximate steady-state values read from these traces are:

Table 5.1: Approximate sensitivity results from the 230 dB comparison.

Modification	Transformer	P-SSHI
Nominal match	1.2 mW	0.07 mW
C_p reduced by 75 nF	55 μW	330 μW
L_p reduced by 1 H	1.0 mW	0.08 mW
L_p reduced by 10 H	36 μW	72 μW

The non-monotonic behavior is expected near a high-Q resonance: a one-henry change is small relative to 42.7 H but can already move the source away from exact cancellation, while a ten-henry shift is enough to remove most of the benefit.

5.7 Frequency and Winding-Resistance Sensitivity

Other LTspice sweeps quantified the same effect at lower source levels. With a nominal C_p-matched primary at 42 Hz, one diode/transformer configuration produced about 32.5 nW.

Changing the source to 42.1 Hz reduced the result to about 10 nW, and 43 Hz reduced it to about 0.15 nW. Adding 200 Ω primary and 100 Ω secondary winding resistance reduced the 42 Hz result to about 1.6 nW.

At 100 Hz, a 7.5 H test primary gave about 200 nW; a 1 Hz frequency error reduced that result to about 8 nW. Adding 20 Ω/10 Ω winding resistance reduced it to about 53 nW. These sweeps explain why a mathematically matched ideal transformer can look exceptional while an actual catalog transformer performs much worse.

5.8 Real Transformer Candidates and Reactive Capacitor Tuning

A high-inductance transformer close to the 42 Hz target was modeled with approximately 44 H on both windings and about 178 Ω winding resistance. It produced roughly 1.15 nW in the tested 42 Hz case, compared with roughly 160 nW for an exact idealized match. At 200 Hz, Hammond 560J winding configurations produced tens of nanowatts with their measured winding resistances, while idealized matching could reach substantially higher values.

Because exact inductance is difficult to obtain with a fixed catalog part, a parallel capacitor bank was also tested as a tuning variable. If a fixed transformer inductance is lower than the ideal matched inductance, the required total capacitance is

$$C_{\text{total}} = \frac{1}{\omega^2 L_p}, \qquad C_b = C_{\text{total}} - C_p. \tag{5.9}$$

In one 200 Hz example, adding approximately 177 nF increased the simulated result from roughly 1.2 pW without the added capacitor to approximately 36.6 nW. A ±10 nF error caused about a threefold reduction, while single-nanofarad adjustments produced much smaller changes. This result motivated the switched coarse/fine capacitor bank in the proposed PCB design.

5.9 Diode Selection in the Low-Voltage Case

At the lowest underwater source voltage, diode forward voltage and leakage both became significant. A low-forward-drop Schottky diode was not automatically best because leakage could consume current after voltage transformation. For example, in one transformed high-impedance case, a low-leakage silicon diode outperformed BAT54 despite the latter's lower forward drop. In another broader sweep, MBR0520L gave the best result at a particular turns ratio. These observations are treated as source-specific component-selection results, not a general conclusion that one diode technology is always superior.

Chapter 6

Multiple Piezoelectric Elements and Phase Mismatch

6.1 Why Series/Parallel Connection Was Tested

Combining multiple piezoelectric elements is an attractive way to raise voltage or current. If sources are in phase, series connection can increase voltage and parallel connection can increase current. In practice, different mechanical locations or resonance frequencies can introduce phase mismatch, so direct AC combination can cause cancellation.

Du et al. experimentally studied this problem for two piezoelectric cantilevers under noisy excitation [6]. They measured about 8.4 µW when each harvester used its own full-bridge rectifier and both rectifier outputs charged a common storage capacitor. Direct parallel connection before one bridge produced about 3.3 µW, and direct series connection produced about 2.7 µW. Their conclusion was that phase mismatch can make raw series/parallel connection substantially worse than separately rectifying each harvester.

6.2 Simulated Series and Phase-Shift Tests

The LTspice model was duplicated to test the same mechanism. With two identical in-phase sources connected in series ahead of the SEH bridge, the source voltages added and the output increased. Reversing one source caused strong cancellation, and imposing a 90-degree phase difference reduced the benefit relative to the in-phase case.

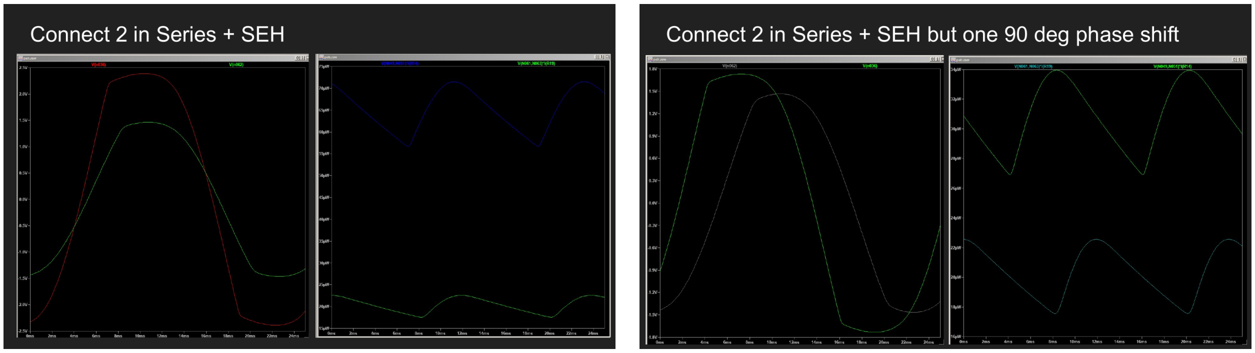


Figure 6.1: Two-source series simulations: in-phase connection (left) and a 90-degree phase mismatch (right).

The alternative was to rectify each source separately and combine only on the DC side. This prevents one AC source from directly discharging or cancelling another at a different phase.

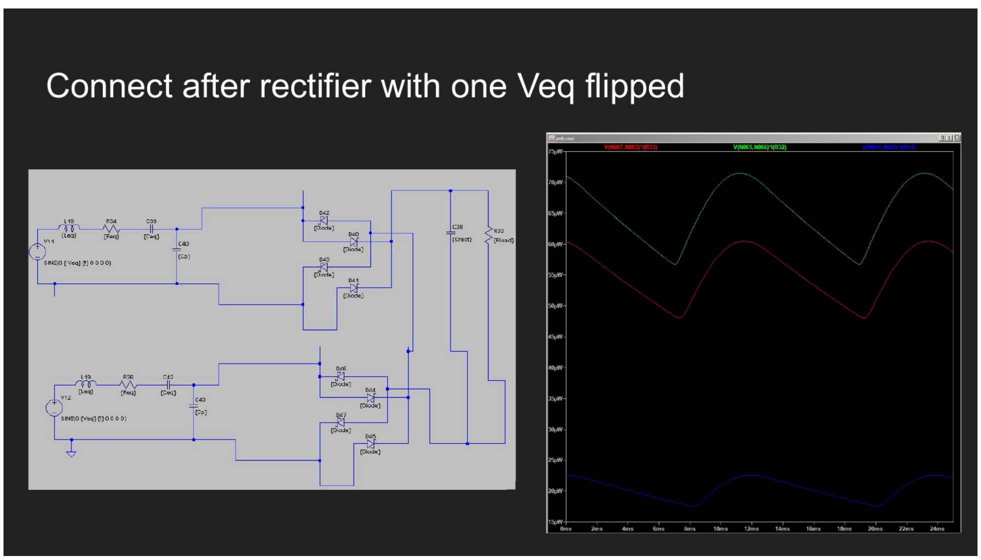


Figure 6.2: Simulation in which two piezoelectric sources are rectified separately and combined at the storage/load node, following the architecture favored by Du et al. under phase mismatch [6].

These results support a modular multi-piezo PCB in which each element can be tested either before or after rectification. The architecture should not assume phase alignment until it has been measured in the final mechanical/acoustic installation.

Chapter 7

Discussion, Hardware Direction, and Future Work

7.1 One Optimization Problem Across Regimes

The simulations form one continuous progression. The vibration benchmark established whether the nonlinear interface and impedance equations were being modeled correctly. Self-powered switching exposed timing loss. The underwater equivalent source then changed the available voltage and impedance, and LTspice exposed component losses that were hidden by ideal Simscape switches. Transformer and multi-source simulations were not separate research questions; they were attempts to solve the same interface problem after the dominant limitation changed.

At conventional benchmark voltage, P-SSHI was clearly the best of the three tested ideal interfaces. At the lowest underwater voltage, semiconductor loss and impedance transformation dominated. At 230 dB, where $V_{oc} \approx 1.73\,\mathrm{V}$, carefully matched transformer coupling could still exceed P-SSHI by a large factor, showing that the advantage was not simply a diode-threshold artifact.

7.2 Why Transformer Matching Is Powerful but Fragile

The transformer result can be understood by separating two operations. Equation (5.4) cancels the source capacitance at the target frequency; Eq. (5.8) then transforms the load resistance. When both are favorable and winding loss is small, the source sees a much more useful electrical load and power can rise sharply. But the same resonance makes the result highly sensitive to component tolerances and excitation frequency. This sensitivity is the main reason the transformer should be treated as a calibrated narrowband interface rather than a universally superior topology.

P-SSHI has the opposite trade-off. It does not require a continuously resonant multi-henry primary and can use a much smaller inductor, but it requires accurate event timing and an efficient switching path. The self-powered BJT experiment showed that a nearly correct LC duration is insufficient if the trigger arrives almost 1 ms late.

7.3 Proposed Modular PCB

The most useful hardware implementation is therefore a modular board rather than one fixed interface. The current design direction is:

1. a piezoelectric input connector and high-impedance test points;
2. selectable SEH and P-SSHI branches so that the baseline and switched interface can be compared on the same source;
3. multiple selectable L_i values for P-SSHI timing tests;
4. a transformer branch that can be bypassed;
5. one or more transformer winding configurations or external transformer connectors;
6. a switched capacitor bank for coarse/fine reactive tuning;
7. interchangeable rectifier diode footprints or daughter boards;
8. an adjustable resistive load for measured load sweeps; and
9. connectors that allow one or multiple piezoelectric elements to be combined before or after rectification.

The first measurements should characterize V_{oc}, source impedance, frequency, and phase before selecting a fixed transformer or capacitor setting.

7.4 Limitations

The reported results are primarily simulation results. The Liang–Liao benchmark is grounded in a published experiment, but the underwater circuit values, transformer sweeps, and component selections still require bench validation with the actual transducer and magnetic parts. Core loss, frequency-dependent inductance, parasitic capacitance, PCB leakage, cable capacitance, and acoustic/mechanical boundary conditions can all move the optimum. The 230 dB transformer/P-SSHI comparison is therefore evidence of a potentially strong matched operating point, not proof that the same ratio will be obtained from an arbitrary physical transformer.

The active zero-cross circuits also need a more systematic timing analysis. Their plotted results show that the tested versions did not beat the passive detector, but this does not show that active switching is intrinsically worse. Published active/self-powered circuits use different sensing and timing strategies [7, 8].

7.5 Future Work

The next steps are intentionally limited to experiments and circuit ideas that were not yet validated in the main results:

1. measure the actual underwater transducer's terminal impedance, open-circuit voltage, dominant frequency content, and phase between multiple elements;
2. test the transformer and capacitor bank with measured rather than nominal inductance and DCR;
3. reproduce the 230 dB transformer/P-SSHI comparison on hardware and sweep intentional detuning around the matched point;
4. investigate better zero-cross/peak-timing circuits in more detail, including strategies from Zouari et al. and Wu et al.;
5. test active rectification, MOSFET bridge approaches, and SECE as later interface alternatives only after their quiescent power and leakage are included;
6. design and test a cold-start/energy-reservoir path if active circuitry is required at the lowest source level; and
7. compare direct series/parallel piezoelectric arrays with individual rectifiers under measured phase mismatch.

Chapter 8

Conclusions

The work answered the research question by building the interface model in layers and checking each layer against a more physical constraint. In the vibration benchmark, the simulation reproduced both impedance and power, not merely waveform appearance. The final maxima were about 0.49 mW for SEH, 1.75 mW for S-SSHI, and 2.18 mW for P-SSHI. Eq. 42 predicted the representative P-SSHI harvested power within about 0.2%, and Eq. 43 predicted the fundamental extracted power within about 3%.

Replacing ideal timing with a self-powered BJT detector reduced the P-SSHI maximum to about 1.82 mW. The measured switch duration was close to the expected half-LC interval, but the approximately 950 µs trigger delay showed why switching phase is a separate design variable from LC duration.

For the underwater transducer, the old and new Lampe equivalent circuits were numerically interchangeable at the terminal level, so the important quantities were the resulting V_{oc}, source impedance, and frequency rather than the internal representation. At 180 dB the source was only about 5.46 mV; at 230 dB it was about 1.73 V.

The LTspice stage then showed that transformer matching can be extremely powerful when the primary inductance satisfies the reactive match and winding loss is low. At 230 dB, one tuned transformer simulation produced roughly 1.2 mW compared with roughly 0.07 mW for the compared P-SSHI configuration. However, detuning L_p, changing C_p, or adding winding resistance could reduce the output by orders of magnitude. This is why magnetic-transformer matching is less common than SSHI in low-frequency piezoelectric harvesters despite its high simulated peak performance: it is narrowband and demands a large, low-loss primary inductance.

The most defensible conclusion is therefore conditional. P-SSHI is the strongest validated nonlinear interface in the conventional vibration benchmark. Transformer/reactive matching can be stronger for a stable underwater source when the source capacitance, frequency, magnetizing inductance, turns ratio, and winding resistance are all controlled. A practical interface should therefore be tunable and modular, and its final selection should follow measurement of the actual transducer rather than assumption of a single universally optimal topology.

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